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The relationship between serum insulin-like growth factor 1 levels and ischemic stroke: a systematic review and meta-analysis

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The relationship between serum insulin-like growth factor 1 levels and ischemic stroke: a systematic review and meta-analysis

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Abbreviations

IGF-1: insulin-like growth factor 1; MD: mean difference; CI: confidence interval; IGFs: Insulin-like growth factors; NO: nitric oxide; CNKI: China National Knowledge Infrastructure; CT: computed tomography; MRI: magnetic resonance imaging; BMI: body mass index; NIHSS: National Institutes of Health Stroke Scale; mRS: modified Rankin Scale; NOS: Newcastle-Ottawa Scale; GH: Growth hormone; IGFBPs: insulin-like growth factor binding proteins; IGF-1R: insulin-like growth factor 1 receptor; CSF: cerebral spinal fluid; PRISMA: preferred reporting items for systematic reviews and meta-analyses

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Abstract

Objective: To assess the association of insulin-like growth factor 1 (IGF-1) with the risk of incident ischemic stroke and outcome after ischemic stroke.

Design: A systematic review of primary studies.

Setting: Hospital in Western Sweden, Italy, China and Denmark.

Methods: A search was performed to identify eligible studies in electronic databases (PubMed, Scopus, Embase, China National Knowledge Infrastructure and Web of Science) updated to September 21, 2019. The relevant data were extracted to perform the meta-analysis. Review Manager version 5.2 was used to pool data and calculate the mean difference (MD) and its 95% confidence interval (CI). Heterogeneity, subgroup analysis, sensitivity analysis and publication bias were also performed in this meta-analysis.

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Results: A total of 2277 patients were included in 17 studies. This meta-analysis indicated that higher serum IGF-1 levels were significantly correlated with less risk of ischemic stroke (MD=-45.32 95% CI = [-63.70, -26.94], P < 0.00001, I^2=99%) and better improvement of outcome after ischemic stroke (MD=27.52, 95% CI= [3.89, 51.14], P=0.02, I^2=96%). According to subgroup analysis, Heterogeneity comes from country, sample size, male and the time from symptom onset to blood collection. Sensitivity analysis showed that there was no significant influence of any individual study on the pooled mean difference. The effect of high heterogeneity on result credibility was eliminated when four included studies were merged (MD=-30.32, 95% CI= [-36.52, -24.11], P< 0.00001, I^2=0%). Moreover, no potential publication bias was found in this meta-analysis.

Conclusion: Higher serum IGF-1 was significantly correlated with less risk of ischemic stroke. In view of high degree of heterogeneity, it may need more studies to confirm the prognostic value of serum IGF-1 levels in ischemic stroke and explore the heterogeneity sources.

Keywords: insulin-like growth factor-1, outcome, ischemic stroke, risk.

Strengths and limitations of this study

- This meta-analysis is the first one to investigate relationship between serum IGF-1 levels and ischemic stroke.
- It has comprehensive database that include 17 studies comprising a total of 2277 patients.
- Expand the sample range to avoid publication bias.
- A high degree of heterogeneity may cause the unreliability of the results.
- Begg's test of publication bias was not performed on the relationship between serum IGF-1 and poststroke functional independence due to the insufficient included studies.

Introduction

Insulin-like growth factors (IGFs), confirmed as peptide hormones, share similar structural homology with insulin. Insulin-like growth factor 1 (IGF-1) protects cell from apoptosis and serves as a vital regulator in cell survival [1, 2]. A growing body of evidences has established that IGF-1 influenced vascular function in many different ways, possessing anti-inflammatory properties and inducing vasodilation by promoting nitric oxide (NO) production [3, 4]. Recently, substantial evidences have accrued to demonstrate that the IGF-1 played a significant role in many cerebral diseases, for example, depression in ischemic stroke, cerebrovascular disease and Alzheimer's disease [5-7].

Stroke is a leading cause of death and disability worldwide, especially in the current aging society [8]. Animal studies and observational studies on humans have tried to uncovered an association between the IGF-1 and risk

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of ischemic stroke. However, the association of IGF-1 with the risk of incident ischemic stroke still remains unclear. Seven previous studies exploring the association manifested inconsistent conclusions. On the one hand, five studies showed that IGF-1 possessed a protective of IGF-1 against ischemic stroke [9-13]. On the other hand, no association between incident ischemic stroke and IGF-1 was uncovered in two studies [14, 15]. Moreover, IGF-1 is widely distributed and highly expressed in the nervous system [16], which exerts neuroprotective effects and accelerates proliferation of neurons. Some studies found that circulating serum IGF-1 is correlated to improvement of functional outcome after ischemic stroke and might predict outcome after ischemic stroke [13, 14, 17-19]. Actually, outcome after ischemic stroke relies on complex interactions of multiple factors, and it still needs more studies to uncover whether the association is causative.

In order to figure out these gaps, this meta-analysis was performed to evaluate the association of IGF-1 with the risk of incident ischemic stroke and outcome after ischemic stroke.

Methods

Study protocol

This meta-analysis was performed in accordance with the preferred reporting items for systematic reviews and meta-analyses (PRISMA) statement (Additional file 1).

Literature search strategy

PubMed, Scopus, Embase, China National Knowledge Infrastructure (CNKI) and Web of Science were searched for studies published or presented on or before September 21, 2019. Publications are not limited to English. To achieve maximum sensitivity of the search strategy and to include more extensive studies, Medical Subject Headings (MeSH) terms and the keywords that were retrieved in the fifield of Title/Abstract were used in combination. “IGF-1” was selected in the MeSH term and “Insulin-Like Somatomedin Peptide I”, “Insulin Like Somatomedin Peptide I”, “Somatomedin C”, “IGF-I-SmC”, “IGF-I”, “Insulin Like Growth Factor I” were retrieved in the fifield of Title/Abstract. All the above were connected by “OR”. “Stroke” was selected in the MeSH term. Ultimately, “AND” was used to connect the two parts. All relevant articles were retrieved first, and then irrelevant articles were manually excluded through title and abstract screening and full-text assessment. All work was conducted independently by two authors to avoid selection bias.

Eligibility criteria

Studies were included if they fulfilled the following criteria:

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- Studies were limited to those involving human subjects.
- There were no restrictions on the information of language, year of the study, patient age, imaging criteria, body mass index (BMI) and National Institutes of Health Stroke Scale (NIHSS) score.
- Study populations were not less than 25 cases and not more than 1000 cases.
- Study had a definite inclusion criterion for acute ischemic stroke.
- Patients suffered from stroke confirmed by computed tomography (CT) or magnetic resonance imaging (MRI) before treatment.
- Studies reported the complete outcomes.
- Study used a prospective or retrospective cohort design, or nested case–control design.
- If duplicate studies with an accumulating number of patients or something else were published, only the most complete studies were included in the analysis performed in this study.

Data extraction and quality assessment

All data were extracted independently from the tables and figures of included studies by two authors in an unblinded fashion or adjudicated by the third author in case of disagreement. Moreover, another author took responsibility for the accuracy and reliability of all of the extracted data. The modified Rankin Scale (mRS) is an evaluation tool to assess the clinical poststroke functional independence. For each study, the study characteristics (author, publication year, study period, country,), main characteristics of the patients (age gender and BMI) and outcomes (the number of ischemic stroke patients, the number of people in control, serum IGF-1 and mRS at 90 days) were extracted independently by two authors. The quality of the included studies was assessed by two other authors using the modified Newcastle-Ottawa Scale (NOS), while 6 points or more was considered to be of high quality.

Statistical analyses

Statistical analysis was implemented using Review Manager version 5.2 (The Nordic Cochrane Centre, The Cochrane Collaboration, Copenhagen, Denmark). The combined effect of mean difference (MD) and 95% confidence interval (CI) were defined by a random-effect model or a fixed-effect model. The heterogeneity among included studies was tested using the inconsistency index (I^2). A random effect model was used when there was a high level of heterogeneity (I^2 values > 50%). A P value <0.10 was considered statistically significant. For indicators with higher heterogeneity, sensitivity analysis and subgroup analyses were carried out

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to identify the reliability of the meta-analysis and explore the sources of heterogeneity with the Stata 12.0 software. Publication bias was evaluated qualitatively by visual observation of funnel graphs and was assessed quantitatively by the Begg's test (Stata 12.0), which manifested a publication bias with $P < 0.05$.

Results

Study selection and characteristics

Detailed steps of the inclusion and exclusion criteria of the study are shown in Fig 1. Briefly, a total of 703 studies were retrieved through five electronic database searches. Thirty-eight additional studies were identified from the references of included studies after removing the duplicates. On the basis of irrelevant titles and abstracts, 526 studies were excluded. Several studies were excluded because they were reviews, non-human experiments; did not present the corresponding outcomes and lack usable data. Ultimately, the final meta-analysis was performed after complete evaluation of 17 studies that met the inclusion criteria [10-14, 20-30].The basic characteristics and quality scores of the included studies were presented in Table 1. The quality scores of studies were proved to be acceptable (Table 2).

Fig. 1 Study search and selection flow diagram.

Relationship between serum IGF-1 and ischemic stroke risk

The relationship between serum IGF-1 and ischemic stroke patients was evaluated, and there was high degree of heterogeneity among these included fifteen studies (I^2=99%). Thus, the random-effects model was utilized to pool MDs. The consequence manifested a significant correlation (MD = -45.32, 95% CI = [-63.70, -26.94], $P < 0.00001$, I^2=99%, random effect) (Fig. 2).

Fig. 2 Forest plot of the relationship between serum IGF-1 and ischemic stroke risk.

Relationship between serum IGF-1 and poststroke functional independence

Poststroke functional independence was identified by mRS at 90 days. The relationship between serum IGF-1 and poststroke functional independence was evaluated, and the random-effects model was utilized to pool MDs. The result manifested no significant difference (MD=27.52, 95% CI= [3.89, 51.14], P=0.02, I^2=96%, random effect) (Fig. 3).

Fig. 3 Forest plot of the relationship between serum IGF-1 and poststroke functional Independence.

Subgroup analysis of heterogeneity sources

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In view of enormous heterogeneity, subgroup analysis of heterogeneity sources was performed to explore heterogeneity sources. The heterogeneity of each subgroup decreased modestly in varying degrees. Notably, the heterogeneity of the time from symptom onset to blood collection presented a significant decrease in paralleled with other subgroup results (Table 3).

Table 3 Subgroup analysis of heterogeneity sources

Item	Subgroup	Number of studies	MD	95% CI	P value	Heterogeneity (I^2)
Country	Asian	12	-55.44	[-72.07, -38.82]	< 0.00001	97%
	Non-Asian	3	-2.92	[-32.42, 26.58]	0.85	99%
Sample size	≥100	6	-18.84	[-45.77, 8.08]	0.17	99%
	<100	9	-65.16	[-88.48, -41.83]	< 0.00001	97%
Male	≥60%	6	-65.89	[-107.51, -24.26]	0.02	98%
	<60%	8	-35.21	[-65.91, -4.52]	0.2	99%
The time from symptom onset to blood collection	≤48 hours	7	-30.04	[-40.05, -20.03]	< 0.00001	86%
	>48 hours	4	-52.40	[-120.79, 15.99]	0.13	99%
	≤24 hours	5	-33.76	[-49.51, -18.01]	< 0.0001	89%

Abbreviations: MD, mean difference; CI, confidence interval.

Sensitivity analysis and adjusted relationship between serum IGF-1 and ischemic stroke

Sensitivity analysis was performed by omitting one study by turns, which showed that there was no significant influence of any individual study on the pooled MDs (Fig. 4). Moreover, in pursuit of low degree of heterogeneity and to uncover low degree of heterogeneity sources, the included studies were randomly merged. It was uncovered that heterogeneity reach to the minimum when four included studies were merged (MD=-30.32, 95% CI= [-36.52, -24.11], P< 0.00001, I^2=0%) (Fig.5 A) [10, 13, 21, 23]. In contrast, the other eleven included studies remained high degree of heterogeneity (MD=-50.34, 95% CI= [-72.82, -27.87],

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P<0.00001, I^2=99%) (Fig. 5 B) [11, 12, 14, 22, 24-30]. Taken together, high degree of heterogeneity seemed not to discredit the final merged result.

Fig. 4 Forest plot in the sensitivity analysis of serum IGF-1 and ischemic stroke risk.

Fig. 5 Forest plot of the adjusted relationship between serum IGF-1 and ischemic stroke risk in (A) four included studies and (B) eleven included studies.

Publication bias analysis

According to Begg's test (P = 0.092 > 0.05) and the funnel plot (Fig. 6), no potential publication bias among the 17 included studies was noted in this meta-analysis.

Fig. 6 Begg's funnel plot for publication bias test of the relationship between serum IGF-1 and ischemic stroke risk.

Discussion

This meta-analysis merged 17 studies comprising 2277 patients to evaluate the association of serum IGF-1 with the risk of incident ischemic stroke and the prognostic value of serum IGF-1 in ischemic stroke. The results demonstrated that serum IGF-1 could be utilized to evaluate the risk of incident ischemic stroke (MD = -45.32, 95% CI = [-63.70, -26.94], P < 0.00001, I^2=99%, random effect) and the outcome after ischemic stroke (MD=27.52, 95% CI= [3.89, 51.14], P=0.02, I^2=96%, random effect). The main mechanism is as follows: Serum IGF-1 mainly comes from the liver, a small part comes from the kidneys and peripherals [31]. In body fluids, IGF-1 has a high affinity with IGF binding proteins (IGFBPs). IGFBPs regulate the biological activity and bioavailability of IGF-1, and the intake of serum IGF-1 enters the cerebrospinal fluid through the blood-brain barrier (BBB) [32]. IGF-1 binds to the widely distributed receptor IGF-1 receptor (IGF-1R) in the brain to activate the two main signal pathways of PI3K/Akt and MAPK, thereby inhibiting neuronal apoptosis [33], promoting the survival of oligodendrocytes [34], and protecting the hippocampus [35].

However, there were inevitable high degree of heterogeneity among the included studies in this meta-analysis. In order to make the conclusion more persuasive and scientific, subgroup analysis was performed to explore the heterogeneity sources. As the results suggested, the time from symptom onset to blood collection appeared to be part of the source of heterogeneity, while the heterogeneity decreased modestly in varying degrees. A study reported that stroke severity was related to decreased IGF-1 in rat as well as human, which meant that different stroke severity of recruited patients may also cause heterogeneity [11, 36]. Besides, insufficient follow-up duration may be the heterogeneity sources. Due to the inability to obtain the original data, the heterogeneity from the time from symptom onset to blood collection, follow-up duration and stroke severity of recruited patients

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cannot be explored further. By scrutinizing the included studies, it was found that ischemic stroke patients who were recruited were subjected to the World Health Organization criteria in some included studies [10, 12, 13], whereas others included studies were not subjected to the criteria. In addition to different patient recruitment criteria, the nutritional status of the patient may cause heterogeneity, which can be indirectly reflect by BMI. Many ischemic stroke patients are followed by accelerated catabolism and decreased food intake, which results in the body muscle mass and BMI decrease. The current study cannot rule out the possible effect of malnutrition on serum IGF-1. Moreover, forest plots in the sensitivity analysis (Fig. 4) and of the adjusted relationship between serum IGF-1 and ischemic stroke risk (Fig. 5) both manifest that the exclusion of any included study and high degree of heterogeneity do not affect the final combined result.

Growth hormone (GH) exerts much of biological activity by promoting the production of IGF-1, and IGF-1 can also mediate the effects of GH in turn. The majority of circulating serum IGF-1 are complexed with IGFBPs and acid labile subunits, which prevent IGF-1 from ligating to the insulin receptor or modulate IGF-1 binding to the IGF-1R [37]. It is universally acknowledged that the bioavailability of IGF-1 is regulated by IGFBPs and only unbound or free IGF-1 possesses biological activity. As the body ageing process, IGF-1 decreases sharply. Some human studies demonstrate the effects of circulating serum IGF-1 on cerebrovascular events may be induced by metabolic syndrome [38, 39]. Moreover, patients suffering from growth hormone deficiency, as a result, may have lower circulating IGF-1 levels, which triggered considerably higher risk of cerebrovascular diseases [40]. Notably, it was reversible with growth hormone replacement [41].

Previous studies evaluating the association of IGF-1 with the risk of incident ischemic stroke and the prognostic value of serum IGF-1 in ischemic stroke reported controversial findings. A study performed by Dong et al. (2014) and colleagues suggested that lower IGF-I levels are significantly related to risk of stroke and stroke severity, but IGFBPs levels was not obtained in that biologically active IGF-I was not fully represented [11]. Johnson et al. (2005) [12] found inverse associations between levels of IGF-I and IGFBP-3 and the risk of ischemic stroke. But another study showing no association with incident ischemic stroke [15]. Meanwhile, Aberg et al. (2011) [14] suggested that serum IGF-1 levels was closely related to improvements in the mRS and Scandinavian Stroke Scale tested at 3 and 24 months. However, the widespread variation of sampling may have influence conclusions. Mattlage et al. (2016) [42] reported the potential importance of the change in IGF-1 and IGF-1 ratio to IGFBP-3 during the first week of stroke in understanding recovery. The limitation of the study was that serum IGF-1 levels were not supplemented with the levels of IGF-1 in the cerebral spinal fluid (CSF). Because in ischemic stroke, IGF-1 transfers from the blood to CSF. Nevertheless, another study showed inconsistent conclusion that higher serum IGF-1 levels was modestly associated with better outcome after ischemic stroke [20].

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To our knowledge, this meta-analysis is the first one to investigate relationship between serum IGF-1 levels and ischemic stroke. Shaheen et al. (2018) [43] have found that serum IGF-1 was an independent risk factor for ischemic stroke in Egyptian population which is consisitent with our study. This research has geographic limitations so this meta-analysis included both English and Chinese studies so that publication bias was avoided in a great deal. Aberg et al. (2019) [44] Have proved that levels of serum IGF-1 are associated with near 3-month outcomes. However, the study only included 354 patients so that the sample size is too small. Otherwise, a comprehensive database search was performed in this meta-anlysis, and as a result, 17 studies comprising a total of 2277 patients were included. Despite the shining points of this meta-analysis, its limitations and shortcomings should be emphasized. A high degree of heterogeneity may cause the unreliability of the results. Meanwhile, Begg's test of publication bias was not performed on the relationship between serum IGF-1 and poststroke functional independence due to the insufficient included studies. Ultimately, a data-feasible study [15] was excluded because of oversize cases, ambiguous patient recruitment criteria and no mention to stroke confirmed by imaging examination.

Since most of the serum IGF-1 binds to IGFBPs, the current measured serum IGF level cannot represent the actual biological activity of igf in circulation. It is hoped that the future meta-analysis will use the data obtained by the IGF-1 specific tyrosine kinase receptor activation analysis method (KIRA) [32] as much as possible. This method indirectly measures the actual biological activity by accurately assessing the activity of the official receptor. This meta-analysis significantly analyzed that higher IGF-1 correlated with less risk of ischemic stroke. It provides a basis for in-depth clinical research on the relationship between the level of IGF-1 and neurological score [30] of stroke patients, post-stroke depression [45] dementi [46] and so on. In addition, it is also of great significance to the prevention of stroke (identification of high-risk groups and chemoprevention), early diagnosis, clinical treatment and prognosis.

Conclusions

This meta-analysis analyzed all the available interrelated information in the published studies and manifested that higher serum IGF-1 was significantly correlated with less risk of ischemic stroke. Concerning to the relationship between serum IGF-1 and outcome after ischemic stroke, it may need more studies to confirm the prognostic value of serum IGF-1 levels.

Patient and Public Involvement

This artical didn't involve patients and or public.

Additional file

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Additional file 1: PRISMA checklist. (DOC)

Authors' contributions

YNL* and WNY* are co-first authors. YNL and WNY designed the study. WNY, YSZ and JHL performed the literature search. WNY, YNL and LCZ selected the studies for inclusion in the meta-analysis. SMC and YNL extracted the data. WNY performed the statistical analysis. YNL, WNY and JHL wrote the manuscript. LH and YYZ revised the manuscript. YYZ provided the funding. All authors read and approved the final manuscript.

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Availability of data and materials

All data analyzed during this study are included in this article.

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

There is no conflict of interest in this article.

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Table 1. Basic characteristics of included studies in the meta-analysis

Study	Study period	Country	The time from symptom onset to blood collection	Age (y) [a or b]	Male (%)	BMI (kg/m^2) [a or b]	Follow-up duration	IGF-I (ng/mL)			
								Risk (Mean, SD or IQR, Total)		mRS (Mean, SD, Total)	
								Case	Control	mRS 0-2	mRS 3-6
Aberg et al. (2019) [20]	1998-2003	Western Sweden	≤24 hours	55.3±11	65.12	NA	7 years	NA	NA	151.5, 6.6, 256	148.3, 12.3, 67
Armbrust et al. (2017) [19]	NA	NA	≤24 hours	72.9±9.5	55.1	26.8±4.4	3 months	NA	NA	131.7, 55.7, 315	114.2, 55.3, 89
Denti et al. (2004) [10]	1998.1-2000.6	Italy	≤24 hours	83±7	34.12	25±5	6 months	69, 45, 85	102, 67, 88	NA	NA
Dong et al. (2014) [11]	2011.8-2013.8	China	≤24 hours	66 (57–77)	58.75	23.9(22.6–26.8)	NA	129, 44, 221	140, 34, 200	NA	NA
Aberg et al. (2011) [14]	1998.8-2003.12	Western Sweden	4 days	55.5±0.86	56.65	26.2±0.43	3 months	172, 3, 388	145, 9, 40	NA	NA

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Johnsen et al. (2005) [12]	1993.12-1997.5	Denmark	NA	50-64	61.00	26.5±4.0	NA	107.1, 33.3, 254	112.9, 33.2, 254	NA	NA
Tang et al. (2014) [13]	2012.8-2013.8	China	≤24 hours	72 (62–83)	53.57	25.3(21.8–28.4)	3 months	127, 50, 168	155, 38, 100	114, 29, 113	146, 43, 55
Zhang et al. (2004) [29]	2000.12-2001.6	China	NA	64±11	66.67	NA	NA	99.48, 33.76, 57	179.9, 35.47, 26	NA	NA
Fu et al. (2001) [22]	Not Stated	China	NA	57-74	55.00	NA	NA	205.7, 125.8, 20	472.6, 317.9, 20	NA	NA
Sun et al. (2001) [27]	1998.9-1999.10	China	≤24 hours	67	70.00	26.45±2.51	NA	91, 51 ,30	175, 49 ,30	NA	NA
Chen et al. (2010) [21]	2009.1-2009.12	China	≤24 hours	65.64±7.59	55	NA	NA	73.5, 28 ,40	103.3, 5.8, 20	NA	NA
Tao et al. (2018) [28]	2016.1-2017.3	China	3 days	60.30±9.75	66.67	NA	3 months	106.54, 48.84, 60	199.2, 53.15, 30	145.81, 49.89, 57	85.56, 32.16, 48

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Zhang et al. (2016) [30]	2013.2-2015.9	China	NA	65.2±8.6	56.00	26.28+1.59	3 months	87.92, 25.31, 50	152.21, 29.32, 50	NA	NA
Li et al. (2016) [24]	2014.5-2015.6	China	3 days	51.9±9.6	56.41	23.9±6.1	NA	98.66, 20.91, 39	153.01, 26.35, 43	NA	NA
Geng et al. (2006) [23]	2003.6-2004.1	China	≤48 hours	65.08± 10.93	70.27	NA	NA	153.22, 64.56, 56	196.1, 33.43, 10	NA	NA
Liu et al. (2002) [25]	1999.2-2002.3	China	3 days	63.85±8.74	63.33	27±5	NA	94, 27, 60	185, 30, 30	NA	NA
Peng et al. (2009) [26]	2006.12-2008.6	China	≤48 hours	NA	NA	NA	NA	10.4, 1.6, 22	28.2, 7.8, 30	NA	NA

[a] Median (IQR); [b] Mean (SD).

Abbreviations: BMI, body mass index; NA, not available; IQR, interquartile range; SD, standard deviation.

Table 2 Newcastle-Ottawa Scale for quality assessment

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Study	Exposed cohort	Non-exposed cohort	Ascertainment of exposure	Outcome of interest	Control for factor	Assessment of outcome	Follow-up long enough	Adequacy of follow-up	Total score
Aberg et al. (2019) [20]	*	*	*	*	**	*	*	*	9
Armbrust et al. (2017) [19]	*	*	*		**	*	*	*	8
Denti et al. (2004) [10]	*	*	*		**	*	*	*	8
Dong et al. (2014) [11]	*	*	*		**	*			6
Aberg et al. (2011) [14]	*	*	*		**	*	*	*	8
Johnsen et al. (2005) [12]	*	*	*		**	*			6
Tang et al.	*	*	*		**	*	*	*	8

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(2014) [13]									
Zhang et al. (2004) [29]	*	*	*		**	*			6
Fu et al. (2001) [22]	*	*	*		**	*			6
Sun et al. (2001) [27]	*	*	*		**	*			6
Chen et al. (2010) [21]	*	*	*		**	*	*		7
Tao et al. (2018) [28]	*	*	*		**	*	*	*	8
Zhang et al. (2016) [30]	*	*	*		**	*	*	*	8
Li et al. (2016) [24]	*	*	*		**	*			6

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Geng et al. (2006) [23]	*	*	*		**	*	*		7
Liu et al. (2002) [25]	*	*	*		**	*			6
Peng et al. (2009) [26]	*	*	*		**	*			6

*The study scored one point in the project; **The study scored two points in the project.

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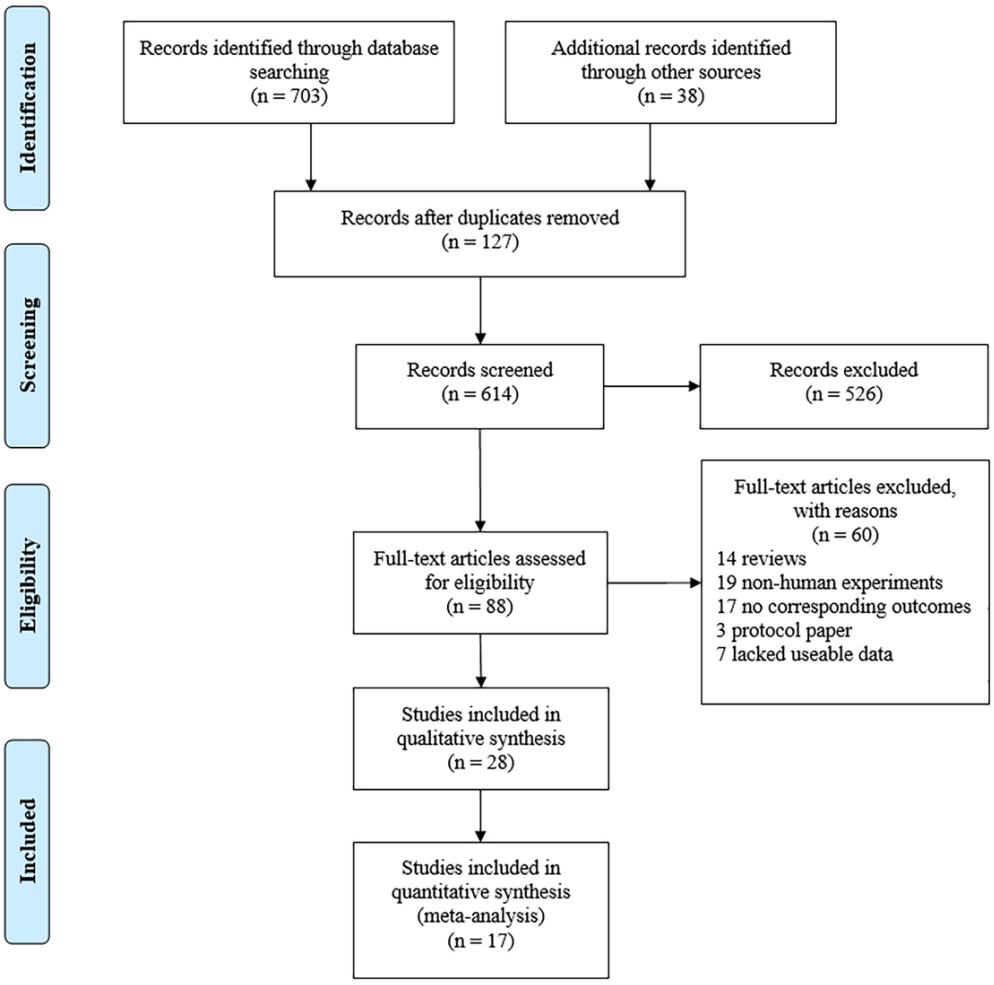


Figure 1

47x46mm (600 x 600 DPI)

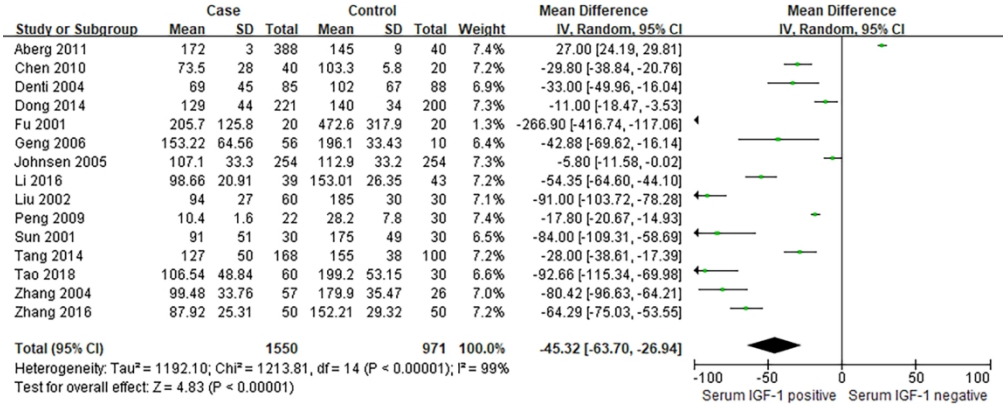


Figure 2

71x29mm (600 x 600 DPI)

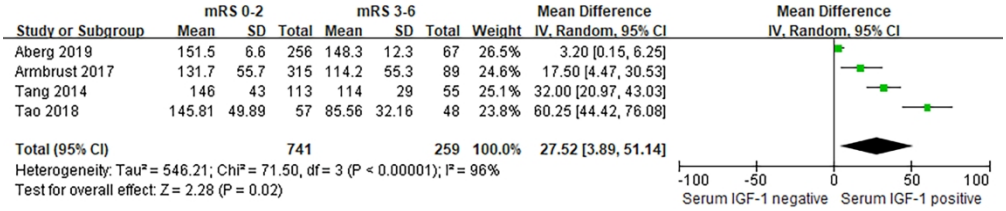


Figure 3

100x21mm (600 x 600 DPI)

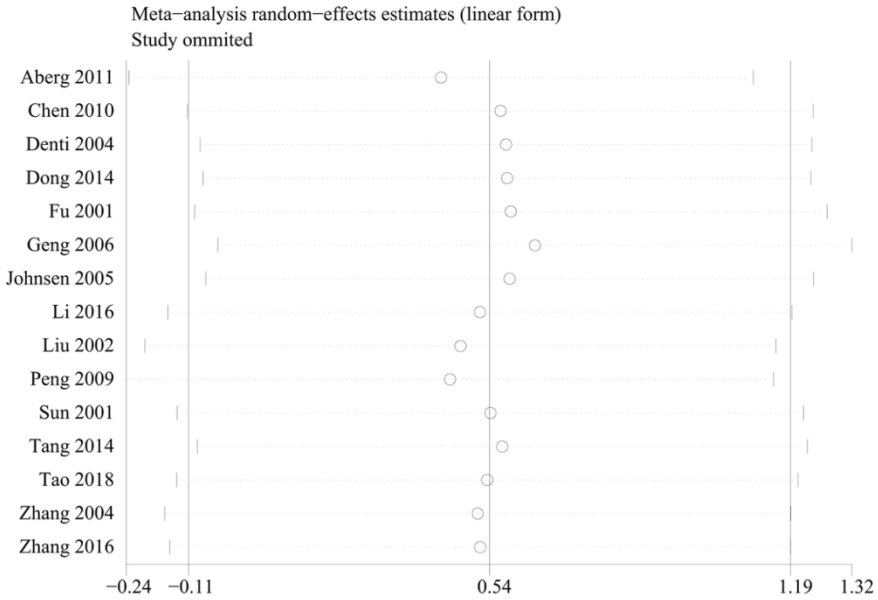


Figure 4

55x37mm (600 x 600 DPI)

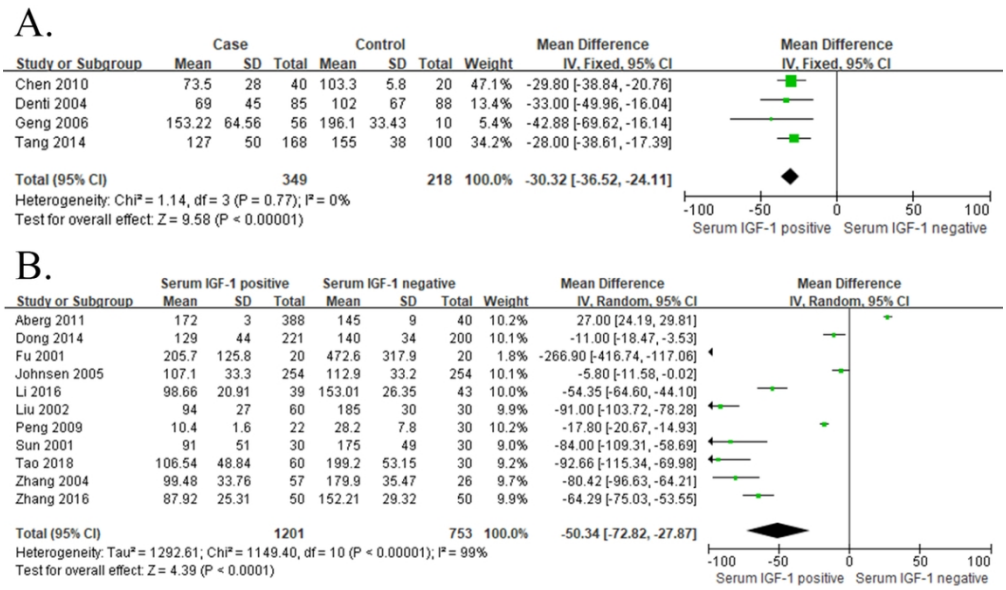


Figure 5

50x30mm (600 x 600 DPI)

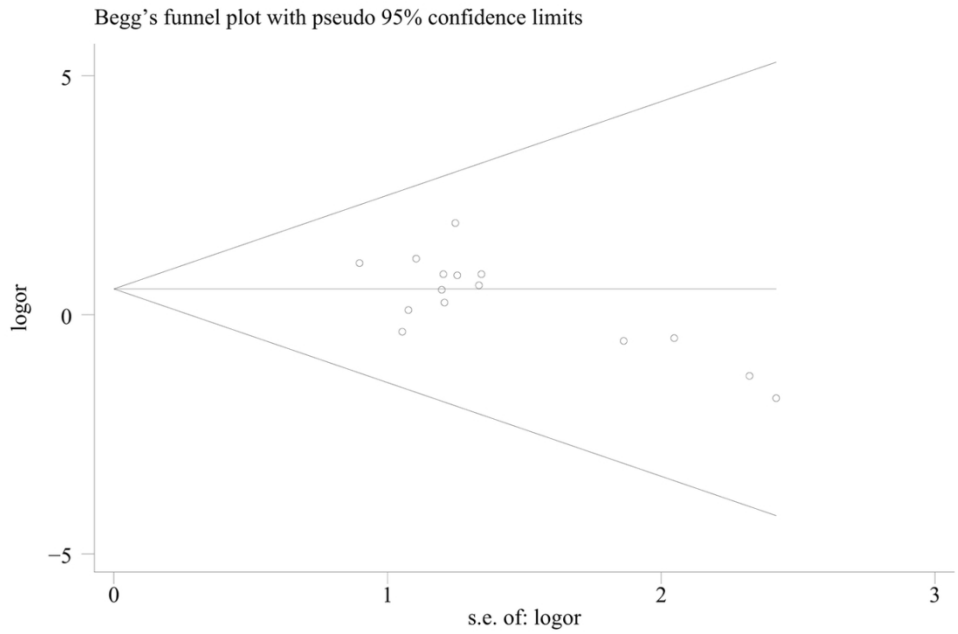
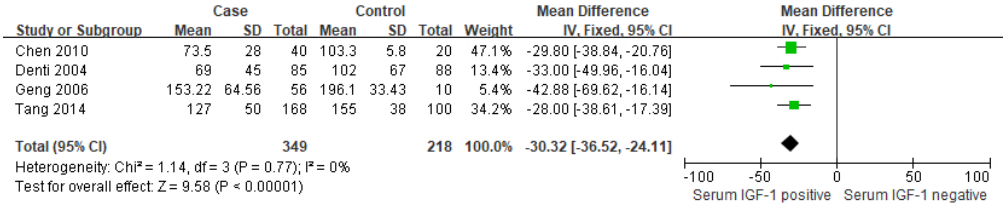


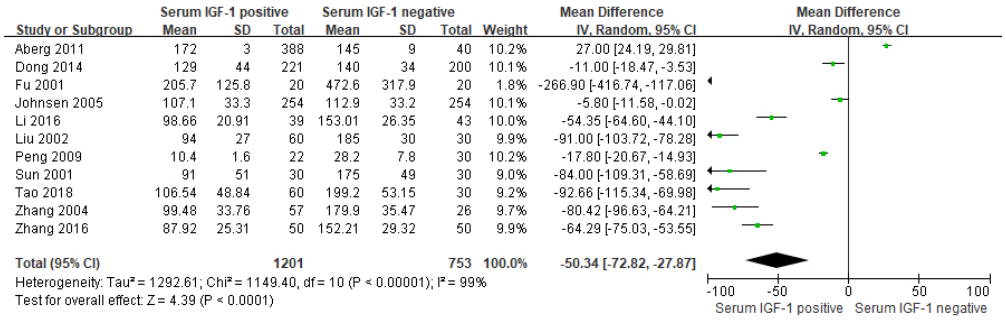
Figure 6

55x36mm (600 x 600 DPI)



4

292x62mm (72 x 72 DPI)



11

315x101mm (72 x 72 DPI)

PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	Title page, page 1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	Abstract, page 2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	Introduction, page 2, 3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	Introduction, page 2, 3
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	Study protocol, page 3
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	Eligibility criteria, page 3
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	Literature search strategy, page 3
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Literature search strategy, page 3
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	Literature search strategy, page 3
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	Data extraction and quality assessment, page 3
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	Data extraction and quality assessment, page 3
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	Data extraction and quality assessment, page 3
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	Statistical analysis, page 4
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	Statistical analysis, page 4

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Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	Statistical analysis, page 4
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	Statistical analysis, page 4
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	Study selection and characteristics, Figure 1, page 4
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	Study selection and characteristics, Table 1, page 4
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	Publication bias analysis and sensitivity analysis, Fig. 6, page 5
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	Fig. 2-5
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	Results, page 4, 5
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	Publication bias analysis and sensitivity analysis, Fig. 6, page 5
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Results, Table 3, Fig 4,5, page 4, 5
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	Discussion, page 6
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	Discussion, page 6, 7
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	Discussion, page 5-7
FUNDING			

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Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	Funding, page 7

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The relationship between serum insulin-like growth factor 1 levels and ischemic stroke: a systematic review and meta-analysis

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Abbreviations

IGF-1: insulin-like growth factor 1; MD: mean difference; CI: confidence interval; IGFs: Insulin-like growth factors; NO: nitric oxide; CNKI: China National Knowledge Infrastructure; CT: computed tomography; MRI: magnetic resonance imaging; BMI: body mass index; NIHSS: National Institutes of Health Stroke Scale; mRS: modified Rankin Scale; NOS: Newcastle-Ottawa Scale; GH: Growth hormone; IGFBPs: insulin-like growth factor binding proteins; IGF-1R: insulin-like growth factor 1 receptor; CSF: cerebral spinal fluid; PRISMA: preferred reporting items for systematic reviews and meta-analyses

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Abstract

Objective: To assess the association of insulin-like growth factor 1 (IGF-1) with the risk of incident ischemic stroke and outcome after ischemic stroke.

Design: A systematic review of primary studies.

Setting: Hospital in Western Sweden, Italy, China and Denmark.

Methods: A search was carried out eligible studies in electronic databases (PubMed, Scopus, Embase, China National Knowledge Infrastructure and Web of Science) updated to December 29, 2020. The relevant data were extracted to perform the meta-analysis. Review Manager version 5.2 was utilized to pool data and calculate the mean difference (MD) and its 95% confidence interval (CI). Heterogeneity, subgroup analysis, sensitivity analysis and publication bias were also performed in this meta-analysis.

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Results: A total of 2277 patients were included in 17 studies. This meta-analysis indicated that higher serum IGF-1 levels were significantly correlated with less risk of ischemic stroke (MD=-45.32 95% CI = [-63.70, -26.94], P < 0.00001, I^2=99%) and better improvement of outcome after ischemic stroke (MD=27.52, 95% CI= [3.89, 51.14], P=0.02, I^2=96%). According to subgroup analysis, Heterogeneity comes from country, sample size, male and the time from symptom onset to blood collection. Sensitivity analysis showed that there was not any significant influence of any individual study on the pooled mean difference. The effect of high heterogeneity on result credibility was eliminated when four included studies were merged (MD=-30.32, 95% CI= [-36.52, -24.11], P< 0.00001, I^2=0%). Moreover, no potential publication bias was discovered in this meta-analysis.

Conclusion: Higher serum IGF-1 was significantly correlated with less risk of ischemic stroke. In view of high degree of heterogeneity, it may need more studies to confirm the prognostic value of serum IGF-1 levels in ischemic stroke and explore the heterogeneity sources.

Keywords: insulin-like growth factor-1, outcome, ischemic stroke, risk.

Strengths and limitations of this study

- This is a meta-analysis of 17 published studies comprising a total of 2277 patients in this study.
- The include studies were heterogeneous in the time from symptom onset to blood collection and follow-up duration at the end of the study.
- Study populations were limited not less than 25 cases and no more than 1000 cases to prevent large samples from affecting the overall results.
- A thorough systematic review and meta-analysis provide confidence in the findings.
- Begg's test of publication bias was not performed on the relationship between serum IGF-1 and poststroke functional independence due to the insufficient included studies.

Introduction

Insulin-like growth factors (IGFs), confirmed as peptide hormones, share similar structural homology with insulin. Insulin-like growth factor 1 (IGF-1) protects cell from apoptosis and serves as a vital regulator in cell survival [1, 2]. A growing body of evidence has established that IGF-1 influenced vascular function in many different ways, possessing anti-inflammatory properties and inducing vasodilation by promoting nitric oxide (NO) production [3, 4]. Recently, substantial evidence has accrued to demonstrate that the IGF-1 played a significant role in many cerebral diseases, for example, depression in ischemic stroke, cerebrovascular disease

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and Alzheimer's disease [5-7].

Stroke is a leading cause of death and disability worldwide, especially in the current aging society [8]. Animal studies and observational studies on humans have been trying to an association between the IGF-1 and risk of ischemic stroke. However, the association of IGF-1 with the risk of incident ischemic stroke still remains unclear. Seven previous studies are exploring the association manifested inconsistent conclusions. On the one hand, five studies showed that IGF-1 possessed a protective effect of IGF-1 against ischemic stroke [9-13]. On the other hand, no association between incident ischemic stroke and IGF-1 was uncovered in two studies [14, 15]. Moreover, IGF-1 is widely dissminated and highly expressed in the nervous system [16], which exerts neuroprotective effects and accelerates proliferation of neurons. Some studies found that circulating serum IGF-1 is correlated with improvement of functional outcome after ischemic stroke and might predict outcome after ischemic stroke [13, 14, 17-19]. Actually, outcome after ischemic stroke relies on complex interactions of multiple factors, and it still needs more studies to disclose whether the association is causative.

In order to figure out these gaps, this meta-analysis was conducted to evaluate the association of IGF-1 with the risk of incident ischemic stroke and outcome after ischemic stroke.

Methods

Study protocol

This meta-analysis was carried out in accordance with the preferred reporting items for systematic reviews and meta-analyses (PRISMA) statement (Additional file 1).

Literature search strategy

PubMed, Scopus, Embase, China National Knowledge Infrastructure (CNKI) and Web of Science were searched for studies published or presented on or before December 29, 2020. Publications are not restricted to English. To achieve maximum sensitivity of the search strategy and to include more extensive studies, Medical Subject Headings (MeSH) terms and the keywords that were retrieved in the field of Title/Abstract were used in combination. "IGF-1" was selected in the MeSH term and "Insulin-Like Somatomedin Peptide I", "Insulin Like Somatomedin Peptide I", "Somatomedin C", "IGF-I-SmC", "IGF-I", "Insulin Like Growth Factor I" were retrieved in the fifield of Title/Abstract. All the above being connected by "OR". "Stroke" was selected in the MeSH term. Ultimately, "AND" was utilized to connect the two parts. All relevant articles were found first, and then irrelevant articles were manually excluded through title and abstract screening and full-text assessment. All work was carried out independently by two authors to avoid selection bias.

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Eligibility criteria

Studies were included if they fulfilled the following criteria:

- Studies were confined to those involving human subjects.
- There were no restrictions on the information of language, year of the study, patient age, imaging criteria, body mass index (BMI) and the National Institutes of Health Stroke Scale (NIHSS) score.
- Study populations were not less than 25 cases and no more than 1000 cases.
- Study had a definite inclusion criterion for acute ischemic stroke.
- Patients suffered from stroke confirmed by computed tomography (CT) or magnetic resonance imaging (MRI) before treatment.
- Studies reported the complete outcomes.
- Study used a prospective or retrospective cohort design, or nested case–control design.
- If duplicate studies with an accumulating number of patients or something else were published, only the most complete studies were included in the analysis performed in this study.

Data extraction and quality assessment

All data were extracted independently from the tables and figures of included studies by two authors in an unblinded fashion or adjudicated by the third author in case of disagreement. Moreover, another author took responsibility for the accuracy and reliability of all of the extracted data. The modified Rankin Scale (mRS) is an evaluation tool to assess the clinical poststroke functional independence. For each study, the study characteristics (author, publication year, study period, country,), main characteristics of the patients (age gender and BMI) and outcomes (the number of ischemic stroke patients, the number of people in control, serum IGF-1 and mRS at 90 days) were extracted independently by two authors. The quality of the included studies was assessed by two other authors using the modified Newcastle-Ottawa Scale (NOS), while 6 points or more was considered to be of high quality.

Outcome measure

The Mean, SD, Total of serum IGF-1 (ng/mL) in case and control groups is a main indicator of stroke risk assessment. The mRS is an evaluation tool to assess the clinical poststroke functional independence (detailed grades of mRS: 0 - No symptoms; 1 - No significant disability; 2 - Slight disability; 3 -Moderate disability; 4 -

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Moderately severe disability; 5 - Severe disability; 6 - Dead). The prespecified primary outcome measures were the mRS score of 0 - 2 and the mRS score 3 - 6.

Statistical analyses

Statistical analysis was implemented using Review Manager version 5.2 (The Nordic Cochrane Centre, the Cochrane Collaboration, Copenhagen, Denmark). The combined effect of mean difference (MD) and 95% confidence interval (CI) were defined by a random-effect model or a fixed-effect model. The heterogeneity among included studies was tested using the inconsistency index (I^2). A random-effect model was used when there was a high level of heterogeneity (I^2 values > 50%). A P value <0.05 was considered statistically significant. For indicators with higher heterogeneity, sensitivity analysis and subgroup analyses were carried out to identify the reliability of the meta-analysis and explore the sources of heterogeneity in the Stata 12.0 software. Publication bias was evaluated qualitatively by visual observation of funnel graphs and was assessed quantitatively by the Begg's test (Stata 12.0), which manifested a publication bias with P <0.05. The number, mean difference (MD), standard deviation (SD) of the experimental group and control group of the included studies and country, sample size, male, the time from symptom onset to blood collection of 17 included studies was imported into Stata 12.0 software. And then select the meta-analysis function to analyze separately.

Results

Study selection and characteristics

Detailed steps of the inclusion and exclusion criteria of the study are shown in Fig 1. Briefly, a total of 703 studies were retrieved through five electronic database searches. Thirty-eight additional studies were identified from the references of included studies after removing the duplicates. On the basis of extraneous titles and abstracts, 526 studies were excluded. Several studies were excluded because they were reviews, non-human experiments; did not present the corresponding outcomes and lack usable data. Ultimately, the final meta-analysis was conducted after complete evaluation of 17 studies that met the inclusion criteria [10-14, 20-30].The basic characteristics and quality score of the included studies were presented in Table 1. The quality score of studies was proved to be acceptable (Table 2).

Fig. 1 Study search and selection flow diagram.

Relationship between serum IGF-1 and ischemic stroke risk

The relationship between serum IGF-1 and ischemic stroke patients was evaluated, and there was a high degree of heterogeneity among these included fifteen studies (I^2=99%). Thus, the random-effects model was

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utilized to pool MDs. The consequence manifested a significant correlation (MD = -45.32, 95% CI = [-63.70, -26.94], P < 0.00001, I^2=99%, random effect) (Fig. 2).

Fig. 2 Forest plot of the relationship between serum IGF-1 and ischemic stroke risk.

Relationship between serum IGF-1 and poststroke functional independence

Poststroke functional independence was identified by mRS at 90 days. The relationship between serum IGF-1 and poststroke functional independence was evaluated, and the random-effects model was utilized to pool MDs. The result manifested no significant difference (MD=27.52, 95% CI= [3.89, 51.14], P=0.02, I^2=96%, random effect) (Fig. 3).

Fig. 3 Forest plot of the relationship between serum IGF-1 and poststroke functional Independence.

Regression of heterogeneity sources

There is high heterogeneity in analyses so we use meta-regression to explore heterogeneity. From the analysis results, we can see that the sample size may be the primary source of heterogeneity (Table 3).

Table 3 Regression of heterogeneity sources

Item	Number of studies	95% CI	P value
Country	15	[-41.50, 78.27]	0.516
Sample size	15	[0.05, 0.28]	0.008
Male	14	[-4.42, 1.57]	0.320
The time from symptom onset to blood collection	14	[-18.71, 83.61]	0.192

Subgroup analysis of heterogeneity sources

In view of enormous heterogeneity, subgroup analysis of heterogeneity sources was performed to explore heterogeneity sources. Heterogeneity of each subgroup decreased modestly in varying degrees. Notably, the heterogeneity of the time from symptom onset to blood collection presented a significant decrease in

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parallel with other subgroup results (Table 4).

Table 4 Subgroup analysis of heterogeneity sources

Item	Subgroup	Number of studies	MD	95% CI	P value	Heterogeneity (I^2)
Country	Asian	12	-55.44	[-72.07, -38.82]	< 0.00001	97%
	Non-Asian	3	-2.92	[-32.42, 26.58]	0.85	99%
Sample size	≥100	6	-18.84	[-45.77, 8.08]	0.17	99%
	<100	9	-65.16	[-88.48, -41.83]	< 0.00001	97%
Male	≥60%	6	-65.89	[-107.51, -24.26]	0.02	98%
	<60%	8	-35.21	[-65.91, -4.52]	0.2	99%
The time from symptom onset to blood collection	≤48 hours	7	-30.04	[-40.05, -20.03]	< 0.00001	86%
	>48 hours	4	-52.40	[-120.79, 15.99]	0.13	99%
	≤24 hours	5	-33.76	[-49.51, -18.01]	< 0.0001	89%

Abbreviations: MD, mean difference; CI, confidence interval.

Sensitivity analysis and adjusted relationship between serum IGF-1 and ischemic stroke

Sensitivity analysis was performed by omitting one study by turns, which showed that there was no significant influence of any individual study on the pooled MDs (Fig. 4). Moreover, in pursuit of low degree of heterogeneity and to uncover low degree of heterogeneity sources, the included studies were randomly merged. It was uncovered that heterogeneity reach to the minimum when four included studies were merged (MD=-30.32, 95% CI= [-36.52, -24.11], P< 0.00001, I^2=0%) (Fig.5 A) [10, 13, 21, 23]. In contrast, the other eleven included studies remained high degree of heterogeneity (MD=-50.34, 95% CI= [-72.82, -27.87], P<0.00001, I^2=99%) (Fig. 5 B) [11, 12, 14, 22, 24-30]. Taken together, high degree of heterogeneity seemed not to discredit the final merged result.

Fig. 4 Forest plot in the sensitivity analysis of serum IGF-1 and ischemic stroke risk.

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Fig. 5 Forest plot of the adjusted relationship between serum IGF-1 and ischemic stroke risk in (A) four included studies and (B) eleven included studies.

Publication bias analysis

According to Begg's test (P = 0.092 > 0.05) and the funnel plot (Fig. 6), no potential publication bias among the 17 included studies was noted in this meta-analysis.

Fig. 6 Begg's funnel plot for publication bias test of the relationship between serum IGF-1 and ischemic stroke risk.

Discussion

The mechanism of serum IGF-1 in evaluating the risk and prognosis of ischemic stroke

This meta-analysis merged 17 studies comprising 2277 patients to evaluate the association of serum IGF-1 with the risk of incident ischemic stroke and the prognostic value of serum IGF-1 in ischemic stroke. The results demonstrated that serum IGF-1 could be utilized to evaluate the risk of incident ischemic stroke (MD = -45.32, 95% CI = [-63.70, -26.94], P < 0.00001, I^2=99%, random-effect) and the outcome after ischemic stroke (MD=27.52, 95% CI= [3.89, 51.14], P=0.02, I^2=96%, random-effect). The main mechanism is as follows: Serum IGF-1 mainly comes from the liver, a small part comes from the kidneys and peripherals [31]. In body fluids, IGF-1 has a high affinity with IGF binding proteins (IGFBPs). IGFBPs regulate the biological activity and bioavailability of IGF-1, and the intake of serum IGF-1 enters the cerebrospinal fluid through the blood-brain barrier (BBB) [32]. IGF-1 binds to the widely distributed receptor IGF-1 receptor (IGF-1R) in the brain to activate the two main signal pathways of PI3K/Akt and MAPK, thereby inhibiting neuronal apoptosis [33], promoting the survival of oligodendrocytes [34], and protecting the hippocampus [35].

Discussion on the causes of heterogeneity

However, there was inevitable high degree of heterogeneity among the included studies in this meta-analysis. In order to make the conclusion more persuasive and scientific, subgroup analysis was performed to explore the heterogeneity sources. As the results suggested, the time from symptom onset to blood collection appeared to be part of the source of heterogeneity, while heterogeneity decreased modestly in varying degrees. A study reported that stroke severity was related to decreased IGF-1 in rat as well as human, which meant that different stroke severity of recruited patients may also cause heterogeneity [11, 36]. Besides, insufficient follow-up duration may be the heterogeneity sources. Due to the inability to obtain the original data, the heterogeneity from the time

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from symptom onset to blood collection, follow-up duration and stroke severity of recruited patients cannot be explored further. By scrutinizing the included studies, it was found that ischemic stroke patients who were recruited were subjected to the World Health Organization criteria in some included studies [10, 12, 13], whereas others included studies were not subjected to the criteria. In addition to different patient recruitment criteria, the nutritional status of the patient may cause heterogeneity, which can be indirectly reflected by BMI. Many ischemic stroke patients are followed by accelerated catabolism and decreased food intake, which results in the body muscle mass and BMI decrease. The current study cannot rule out the possible effect of malnutrition on serum IGF-1. Moreover, forest plots in the sensitivity analysis (Fig. 4) and of the adjusted relationship between serum IGF-1 and ischemic stroke risk (Fig. 5) both manifest that the exclusion of any included study and a high degree of heterogeneity do not affect the final combined result.

The relationship between growth hormone and IGF-1 in cerebrovascular diseases

Growth hormone (GH) exerts much of biological activity by promoting the production of IGF-1, and IGF-1 can also mediate the effects of GH in turn. The majority of circulating serum IGF-1 are completed with IGFBPs and acid labile subunits, which prevent IGF-1 from lighting to the insulin receptor or modulate IGF-1 binding to the IGF-1R [37]. It is universally acknowledged that the bioavailability of IGF-1 is regulated by IGFBPs and only unbound or free IGF-1 possesses biological activity. As the body ageing process, IGF-1 decreases sharply. Some human studies demonstrate the effects of circulating serum IGF-1 in cerebrovascular events may be induced by metabolic syndrome [38, 39]. Moreover, patients suffering from growth hormone deficiency, as a result, may have lower circulating IGF-1 levels, which triggered considerably higher risk of cerebrovascular diseases [40]. Notably, it was reversible with growth hormone replacement [41].

Insufficiency of previous related research

Previous studies assessing the association of IGF-1 with the risk of incident ischemic stroke and the prognostic value of serum IGF-1 in ischemic stroke reported controversial findings. A study performed by Dong et al. (2014) and colleagues suggested that lower IGF-I levels are significantly related to risk of stroke and stroke severity, but IGFBPs levels were not obtained in that biologically active IGF-I was not fully represented [11]. Johnson et al. (2005) [12] found inverse associations between levels of IGF-I and IGFBP-3 and the risk of ischemic stroke. However, another study showing no association with incident ischemic stroke [15]. Meanwhile, Aberg et al. (2011) [14] suggested that serum IGF-1 levels were closely related to improvements in the mRS and Scandinavian Stroke Scale tested at 3 and 24 months. However, the widespread variation of sampling may have to influence conclusions. Mattlage et al. (2016) [42] reported the potential importance of the change in IGF-1

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and IGF-1 ratio to IGFBP-3 during the first week of stroke in understanding recovery. The limitation of the study was that serum IGF-1 levels were not supplemented with the levels of IGF-1 in the cerebral spinal fluid (CSF). Because in ischemic stroke, IGF-1 transfers from the blood to CSF. Nevertheless, another study showed inconsistent conclusion that higher serum IGF-1 levels were modestly associated with better outcome after ischemic stroke [20].

Strengths and limitations

To our knowledge, this meta-analysis is the first one to investigate relationship between serum IGF-1 levels and ischemic stroke. Shaheen et al. (2018) [43] have found that serum IGF-1 was an independent risk factor for ischemic stroke in Egyptian population which is consisitent with our study. This research has geographic limitations so this meta-analysis included both English and Chinese studies so that publication bias was avoided in a great deal. Besides, the eligibility criteria that study populations were not less than 25 cases and not more than 1000 cases to prevent large samples from affecting the overall results. Aberg et al. (2019) [44] have proved that levels of serum IGF-1 are associated with near 3-month outcomes. However, the study only included 354 patients so that the sample size is too small. Otherwise, a comprehensive database search was performed in this meta-anlysis, and as a result, 17 studies comprising a total of 2277 patients were included. Despite the shining points of this meta-analysis, its limitations and shortcomings should be emphasized. A high degree of heterogeneity may cause the unreliability of the results. Meanwhile, Begg's test of publication bias was not performed on the relationship between serum IGF-1 and poststroke functional independence due to the insufficient included studies. Ultimately, a data-feasible study [15] was excluded because of oversize cases, ambiguous patient recruitment criteria and no mention to stroke confirmed by imaging examination.

The clinical significance and future prospects of the study

Since most of the serum IGF-1 binds to IGFBPs, the current measured serum IGF level cannot represent the actual biological activity of igf in circulation. It is hoped that the future meta-analysis will use the data obtained by the IGF-1 specific tyrosine kinase receptor activation analysis method (KIRA) [32] as much as possible. This method indirectly measures the actual biological activity by accurately assessing the activity of the official receptor. This meta-analysis significantly analyzed that higher IGF-1 correlated with less risk of ischemic stroke. It provides a basis for in-depth clinical research on the relationship between the level of IGF-1 and neurological score [30] of stroke patients, post-stroke depression [45] dementi [46] and so on. In addition, it is also of great significance to the prevention of stroke (identification of high-risk groups and chemoprevention), early diagnosis, clinical treatment and prognosis.

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Conclusions

This meta-analysis analyzed all the available interrelated information in the published studies and manifested that higher serum IGF-1 was significantly correlated with less risk of ischemic stroke. Concerning to the relationship between serum IGF-1 and outcome after ischemic stroke, it may need more studies to confirm the prognostic value of serum IGF-1 levels.

Competing interests

None declared.

Patient and Public Involvement

This artical didn't involve patients and or public.

Additional file

Additional file 1: PRISMA checklist. (DOC)

Authors' contributions

YNL* and WNY* are co-first authors. YNL and WNY designed the study. WNY, YSZ and JHL performed the literature search. WNY, YNL and LCZ selected the studies for inclusion in the meta-analysis. SMC and YNL extracted the data. WNY performed the statistical analysis. YNL, WNY and JHL wrote the manuscript. LH and YYZ revised the manuscript. YYZ provided the funding. All authors read and approved the final manuscript.

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Availability of data and materials

All data analyzed during this study are included in this article.

Ethics approval and consent to participate

Not applicable.

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Consent for publication

Not applicable.

Competing interests

There is no conflict of interest in this article.

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Not applicable

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Table 1. Basic characteristics of included studies in the meta-analysis

Study	Study period	Country	The time from symptom onset to blood collection	Age (y) [a or b]	Male (%)	BMI (kg/m^2) [a or b]	Follow-up duration	IGF-I (ng/mL)			
								Risk (Mean, SD or IQR, Total)		mRS (Mean, SD, Total)	
								Case	Control	mRS 0-2	mRS 3-6
Fu et al. (2001) [22]	Not Stated	China	NA	57-74	55.00	NA	NA	205.7, 125.8, 20	472.6, 317.9, 20	NA	NA
Sun et al. (2001) [27]	1998.9-1999.10	China	≤24 hours	67	70.00	26.45±2.51	NA	91, 51 ,30	175, 49 ,30	NA	NA
Liu et al. (2002) [25]	1999.2-2002.3	China	3 days	63.85±8.74	63.33	27±5	NA	94, 27, 60	185, 30, 30	NA	NA
Denti et al. (2004) [10]	1998.1-2000.6	Italy	≤24 hours	83±7	34.12	25±5	6 months	69, 45, 85	102, 67, 88	NA	NA
Zhang et al. (2004) [29]	2000.12-2001.6	China	NA	64±11	66.67	NA	NA	99.48, 33.76, 57	179.9, 35.47, 26	NA	NA

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Johnsen et al. (2005) [12]	1993.12-1997.5	Denmark	NA	50-64	61.00	26.5±4.0	NA	107.1, 33.3, 254	112.9, 33.2, 254	NA	NA
Geng et al. (2006) [23]	2003.6-2004.1	China	≤48 hours	65.08± 10.93	70.27	NA	NA	153.22, 64.56, 56	196.1, 33.43, 10	NA	NA
Peng et al. (2009) [26]	2006.12-2008.6	China	≤48 hours	NA	NA	NA	NA	10.4, 1.6, 22	28.2, 7.8, 30	NA	NA
Chen et al. (2010) [21]	2009.1-2009.12	China	≤24 hours	65.64±7.59	55	NA	NA	73.5, 28 ,40	103.3, 5.8, 20	NA	NA
Aberg et al. (2011) [14]	1998.8-2003.12	Western Sweden	4 days	55.5±0.86	56.65	26.2±0.43	3 months	172, 3, 388	145, 9, 40	NA	NA
Dong et al. (2014) [11]	2011.8-2013.8	China	≤24 hours	66 (57–77)	58.75	23.9(22.6–26.8)	NA	129, 44, 221	140, 34, 200	NA	NA
Tang et al. (2014) [13]	2012.8-2013.8	China	≤24 hours	72 (62–83)	53.57	25.3(21.8–28.4)	3 months	127, 50, 168	155, 38, 100	114, 29, 113	146, 43, 55

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Zhang et al. (2016) [30]	2013.2-2015.9	China	NA	65.2±8.6	56.00	26.28+1.59	3 months	87.92, 25.31, 50	152.21, 29.32, 50	NA	NA
Li et al. (2016) [24]	2014.5-2015.6	China	3 days	51.9±9.6	56.41	23.9±6.1	NA	98.66, 20.91, 39	153.01, 26.35, 43	NA	NA
Armbrust et al. (2017) [19]	NA	NA	≤24 hours	72.9±9.5	55.1	26.8±4.4	3 months	NA	NA	131.7, 55.7, 315	114.2, 55.3, 89
Tao et al. (2018) [28]	2016.1-2017.3	China	3 days	60.30±9.75	66.67	NA	3 months	106.54, 48.84, 60	199.2, 53.15, 30	145.81, 49.89, 57	85.56, 32.16, 48
Aberg et al. (2019) [20]	1998-2003	Western Sweden	≤24 hours	55.3±11	65.12	NA	7 years	NA	NA	151.5, 6.6, 256	148.3, 12.3, 67

[a] Median (IQR); [b] Mean (SD).

Abbreviations: BMI, body mass index; NA, not available; IQR, interquartile range; SD, standard deviation.

Table 2 Newcastle-Ottawa Scale for quality assessment

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Study	Exposed cohort	Non-exposed cohort	Ascertainment of exposure	Outcome of interest	Control for factor	Assessment of outcome	Follow-up long enough	Adequacy of follow-up	Total score
Fu et al. (2001) [22]	*	*	*		**	*			6
Sun et al. (2001) [27]	*	*	*		**	*			6
Liu et al. (2002) [25]	*	*	*		**	*			6
Denti et al. (2004) [10]	*	*	*		**	*	*	*	8
Zhang et al. (2004) [29]	*	*	*		**	*			6
Johnsen et al. (2005) [12]	*	*	*		**	*			6
Geng et al.	*	*	*		**	*	*		7

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(2006) [23]									
Peng et al. (2009) [26]	*	*	*		**	*			6
Chen et al. (2010) [21]	*	*	*		**	*	*		7
Aberg et al. (2011) [14]	*	*	*		**	*	*	*	8
Dong et al. (2014) [11]	*	*	*		**	*			6
Tang et al. (2014) [13]	*	*	*		**	*	*	*	8
Zhang et al. (2016) [30]	*	*	*		**	*	*	*	8
Li et al. (2016) [24]	*	*	*		**	*			6

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Armbrust et al. (2017) [19]	*	*	*		**	*	*	*	8
Tao et al. (2018) [28]	*	*	*		**	*	*	*	8
Aberg et al. (2019) [20]	*	*	*	*	**	*	*	*	9

*The study scored one point in the project; **The study scored two points in the project.

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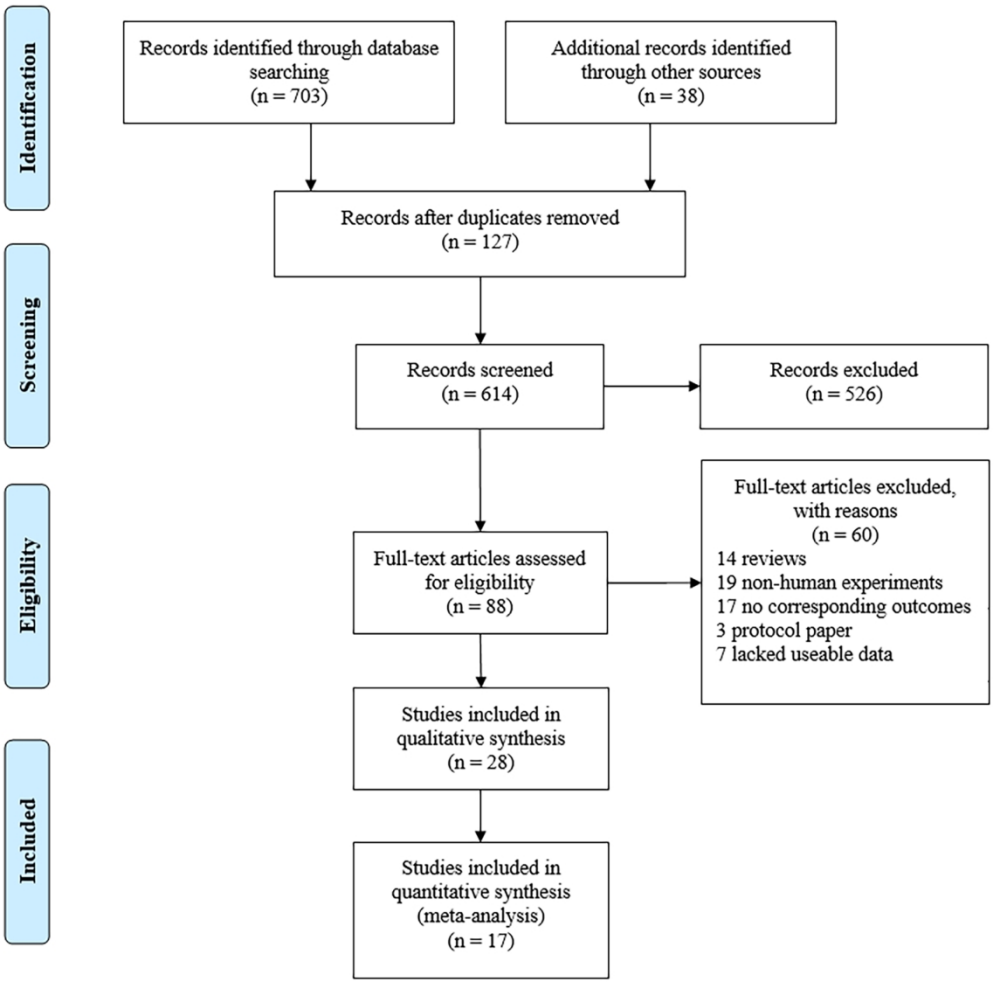


Figure 1

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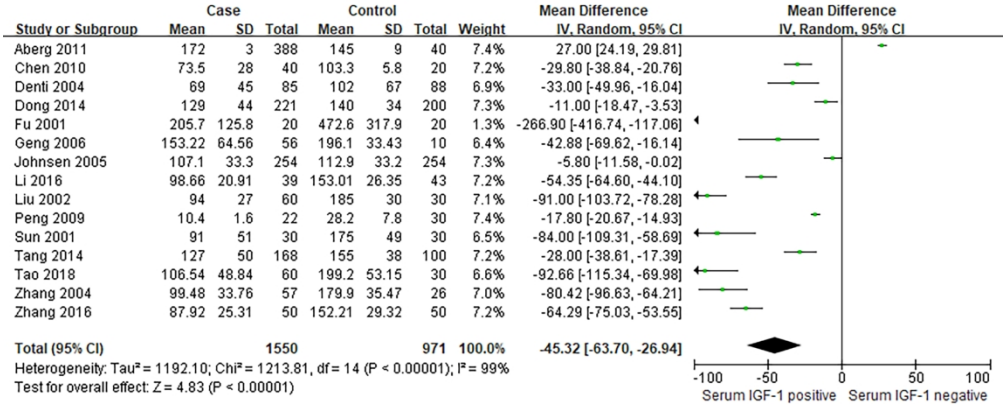


Figure 2

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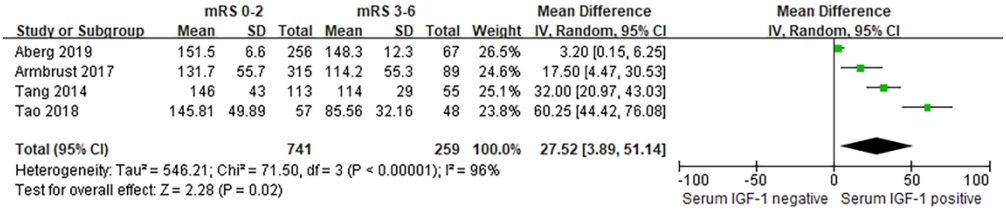


Figure 3

100x21mm (1000 x 1000 DPI)

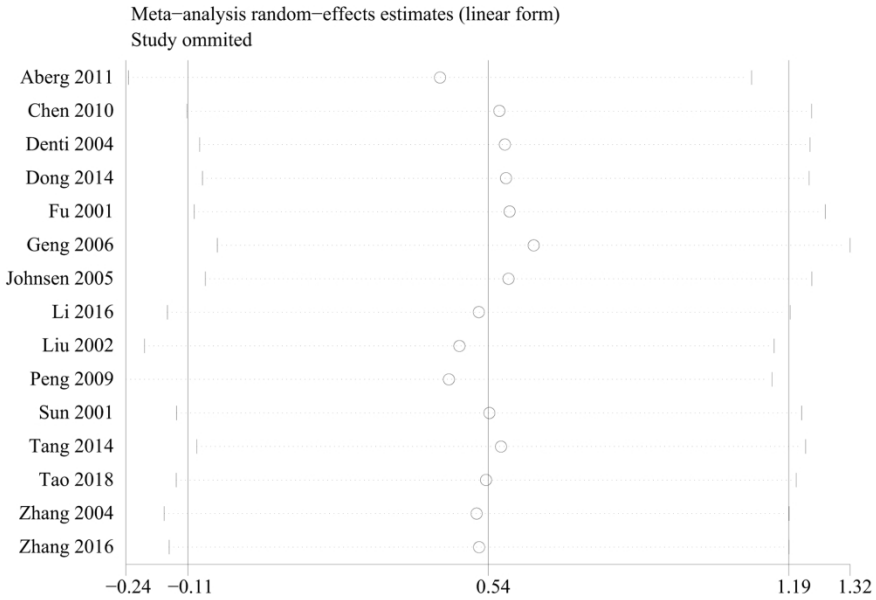


Figure 4

55x37mm (1000 x 1000 DPI)

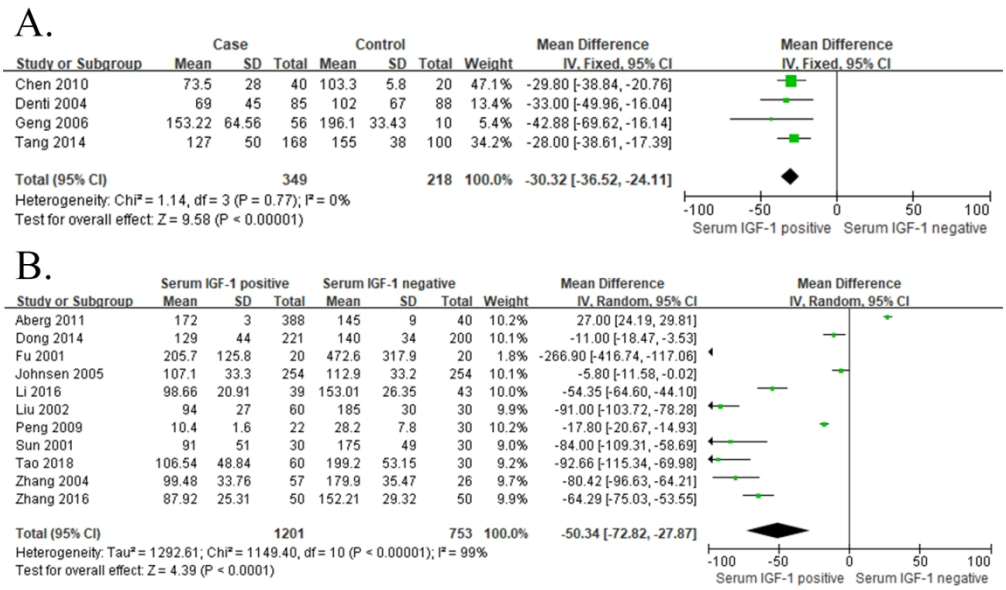


Figure 5

50x30mm (1000 x 1000 DPI)

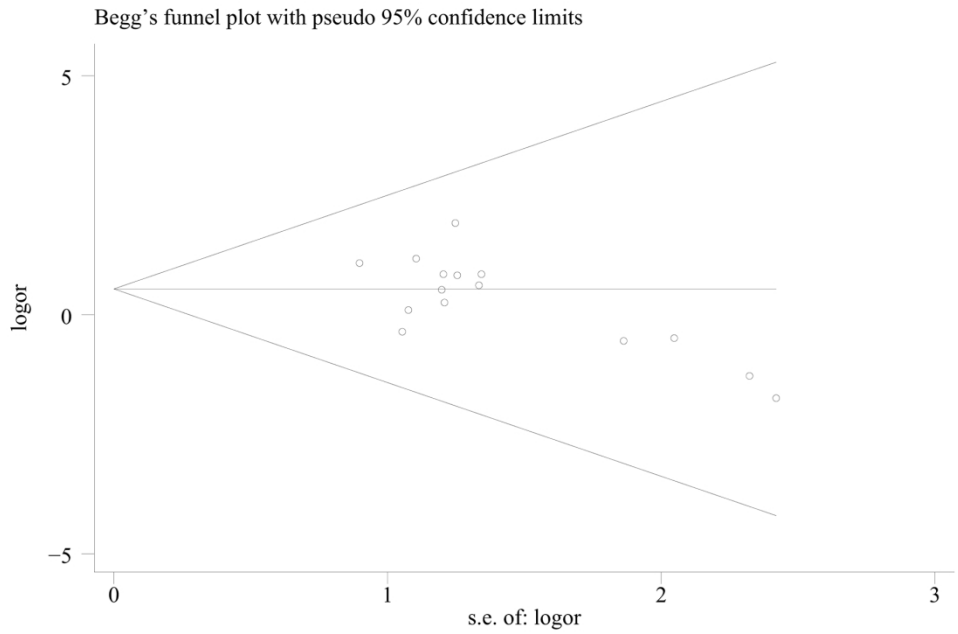

Begg's funnel plot with pseudo 95% confidence limits
5
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Figure 6

PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	Title page, page 1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	Abstract, page 2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	Introduction, page 2, 3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	Introduction, page 2, 3
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	Study protocol, page 3
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	Eligibility criteria, page 3
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	Literature search strategy, page 3
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Literature search strategy, page 3
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	Literature search strategy, page 3
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	Data extraction and quality assessment, page 3
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	Data extraction and quality assessment, page 3
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	Data extraction and quality assessment, page 3
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	Statistical analysis, page 4
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	Statistical analysis, page 4

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	Statistical analysis, page 4
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	Statistical analysis, page 4
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	Study selection and characteristics, Figure 1, page 4
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	Study selection and characteristics, Table 1, page 4
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	Publication bias analysis and sensitivity analysis, Fig. 6, page 5
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	Fig. 2-5
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	Results, page 4, 5
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	Publication bias analysis and sensitivity analysis, Fig. 6, page 5
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Results, Table 3, Fig 4,5, page 4, 5
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	Discussion, page 6
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	Discussion, page 6, 7
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	Discussion, page 5-7
FUNDING			

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PRISMA 2009 Checklist

Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	Funding, page 7

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The relationship between serum insulin-like growth factor 1 levels and ischemic stroke: a systematic review and meta-analysis

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Abbreviations

IGF-1: insulin-like growth factor 1; MD: mean difference; CI: confidence interval; IGFs: Insulin-like growth factors; NO: nitric oxide; CNKI: China National Knowledge Infrastructure; CT: computed tomography; MRI: magnetic resonance imaging; BMI: body mass index; NIHSS: National Institutes of Health Stroke Scale; mRS: modified Rankin Scale; NOS: Newcastle-Ottawa Scale; GH: Growth hormone; IGFBPs: insulin-like growth factor binding proteins; IGF-1R: insulin-like growth factor 1 receptor; CSF: cerebral spinal fluid; PRISMA: preferred reporting items for systematic reviews and meta-analyses

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Abstract

Objective: To assess the association of insulin-like growth factor 1 (IGF-1) with the risk of incident ischemic stroke and outcome after ischemic stroke.

Design: A systematic review of primary studies.

Setting: Hospital in Western Sweden, Italy, China and Denmark.

Methods: A search was carried out eligible studies in electronic databases (PubMed, Scopus, Embase, China National Knowledge Infrastructure and Web of Science) updated to December 29, 2020. The relevant data were extracted to perform the meta-analysis. Review Manager version 5.2 was utilized to pool data and calculate the mean difference (MD) and its 95% confidence interval (CI). Heterogeneity, subgroup analysis, sensitivity analysis and publication bias were also performed in this meta-analysis.

Results: A total of 2277 patients were included in 17 studies. This meta-analysis indicated that higher serum IGF-1 levels were significantly correlated with less risk of ischemic stroke (MD=-45.32 95% CI = [-63.70, -26.94], P < 0.00001, I^2=99%) and better improvement of outcome after ischemic stroke (MD=27.52, 95% CI= [3.89, 51.14], P=0.02, I^2=96%). According to subgroup analysis, Heterogeneity comes from country, sample size, male and the time from symptom onset to blood collection. Sensitivity analysis showed that there was not any significant influence of any individual study on the pooled mean difference. The effect of high heterogeneity on result credibility was eliminated when four included studies were merged (MD=-30.32, 95% CI= [-36.52, -24.11], P< 0.00001, I^2=0%). Moreover, no potential publication bias was discovered in this meta-analysis.

Conclusion: Higher serum IGF-1 was significantly correlated with less risk of ischemic stroke. In view of high degree of heterogeneity, it may need more studies to confirm the prognostic value of serum IGF-1 levels in ischemic stroke and explore the heterogeneity sources.

Keywords: insulin-like growth factor-1, outcome, ischemic stroke, risk.

Strengths and limitations of this study

- This is a meta-analysis of 17 published studies comprising a total of 2277 patients in this study.
- The include studies were heterogeneous in the time from symptom onset to blood collection and follow-up duration at the end of the study.
- Study populations were limited not less than 25 cases and no more than 1000 cases to prevent large samples from affecting the overall results.
- A thorough systematic review and meta-analysis provide confidence in the findings.
- Begg's test of publication bias was not performed on the relationship between serum IGF-1 and poststroke functional independence due to the insufficient included studies.

Introduction

Insulin-like growth factors (IGFs), confirmed as peptide hormones, share similar structural homology with insulin. Insulin-like growth factor 1 (IGF-1) protects cell from apoptosis and serves as a vital regulator in cell survival [1, 2]. A growing body of evidence has established that IGF-1 influenced vascular function in many

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different ways, possessing anti-inflammatory properties and inducing vasodilation by promoting nitric oxide (NO) production [3, 4]. Recently, substantial evidence has accrued to demonstrate that the IGF-1 played a significant role in many cerebral diseases, for example, depression in ischemic stroke, cerebrovascular disease and Alzheimer's disease [5-7].

Stroke is a leading cause of death and disability worldwide, especially in the current aging society [8]. Animal studies and observational studies on humans have been trying to an association between the IGF-1 and risk of ischemic stroke. However, the association of IGF-1 with the risk of incident ischemic stroke still remains unclear. Seven previous studies are exploring the association manifested inconsistent conclusions. On the one hand, five studies showed that IGF-1 possessed a protective effect of IGF-1 against ischemic stroke [9-13]. On the other hand, no association between incident ischemic stroke and IGF-1 was uncovered in two studies [14, 15]. Moreover, IGF-1 is widely dissminated and highly expressed in the nervous system [16], which exerts neuroprotective effects and accelerates proliferation of neurons. Some studies found that circulating serum IGF-1 is correlated with improvement of functional outcome after ischemic stroke and might predict outcome after ischemic stroke [13, 14, 17-19]. Actually, outcome after ischemic stroke relies on complex interactions of multiple factors, and it still needs more studies to disclose whether the association is causative.

In order to figure out these gaps, this meta-analysis was conducted to evaluate the association of IGF-1 with the risk of incident ischemic stroke and outcome after ischemic stroke.

Methods

Patient and Public Involvement

This artical didn't involve patients and or public.

Study protocol

This meta-analysis was carried out in accordance with the preferred reporting items for systematic reviews and meta-analyses (PRISMA) statement (Additional file 1).

Literature search strategy

PubMed, Scopus, Embase, China National Knowledge Infrastructure (CNKI) and Web of Science were searched for studies published or presented on or before December 29, 2020. Publications are not restricted to English. To achieve maximum sensitivity of the search strategy and to include more extensive studies, Medical Subject Headings (MeSH) terms and the keywords that were retrieved in the field of Title/Abstract were used in combination. "IGF-1" was selected in the MeSH term and "Insulin-Like Somatomedin Peptide I", "Insulin Like Somatomedin Peptide I", "Somatomedin C", "IGF-I-SmC", "IGF-I", "Insulin Like Growth Factor I" were retrieved in the fifield of Title/Abstract. All the above being connected by "OR". "Stroke" was selected in the MeSH term. Ultimately, "AND" was utilized to connect the two parts. All relevant articles were found first, and

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then irrelevant articles were manually excluded through title and abstract screening and full-text assessment. All work was carried out independently by two authors to avoid selection bias.

Eligibility criteria

Studies were included if they fulfilled the following criteria:

- Studies were confined to those involving human subjects.
- There were no restrictions on the information of language, year of the study, patient age, imaging criteria, body mass index (BMI) and the National Institutes of Health Stroke Scale (NIHSS) score.
- Study populations were not less than 25 cases and no more than 1000 cases.
- Study had a definite inclusion criterion for acute ischemic stroke.
- Patients suffered from stroke confirmed by computed tomography (CT) or magnetic resonance imaging (MRI) before treatment.
- Studies reported the complete outcomes.
- Study used a prospective or retrospective cohort design, or nested case–control design.
- If duplicate studies with an accumulating number of patients or something else were published, only the most complete studies were included in the analysis performed in this study.

Data extraction and quality assessment

All data were extracted independently from the tables and figures of included studies by two authors in an unblinded fashion or adjudicated by the third author in case of disagreement. Moreover, another author took responsibility for the accuracy and reliability of all of the extracted data. The modified Rankin Scale (mRS) is an evaluation tool to assess the clinical poststroke functional independence. For each study, the study characteristics (author, publication year, study period, country,), main characteristics of the patients (age gender and BMI) and outcomes (the number of ischemic stroke patients, the number of people in control, serum IGF-1 and mRS at 90 days) were extracted independently by two authors. The quality of the included studies was assessed by two other authors using the modified Newcastle-Ottawa Scale (NOS), while 6 points or more was considered to be of high quality.

Outcome measure

The Mean, SD, Total of serum IGF-1 (ng/mL) in case and control groups is a main indicator of stroke risk assessment. The mRS is an evaluation tool to assess the clinical poststroke functional independence (detailed grades of mRS: 0 - No symptoms; 1 - No significant disability; 2 - Slight disability; 3 -Moderate disability; 4 - Moderately severe disability; 5 - Severe disability; 6 - Dead). The prespecified primary outcome measures were the mRS score of 0 - 2 and the mRS score 3 - 6.

Statistical analyses

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Statistical analysis was implemented using Review Manager version 5.2 (The Nordic Cochrane Centre, the Cochrane Collaboration, Copenhagen, Denmark). The combined effect of mean difference (MD) and 95% confidence interval (CI) were defined by a random-effect model or a fixed-effect model. The heterogeneity among included studies was tested using the inconsistency index (I^2). A random-effect model was used when there was a high level of heterogeneity (I^2 values > 50%). A P value <0.05 was considered statistically significant. For indicators with higher heterogeneity, sensitivity analysis and subgroup analyses were carried out to identify the reliability of the meta-analysis and explore the sources of heterogeneity in the Stata 12.0 software. Publication bias was evaluated qualitatively by visual observation of funnel graphs and was assessed quantitatively by the Begg's test (Stata 12.0), which manifested a publication bias with P <0.05. The number, mean difference (MD), standard deviation (SD) of the experimental group and control group of the included studies and country, sample size, male, the time from symptom onset to blood collection of 17 included studies was imported into Stata 12.0 software.

Results

Study selection and characteristics

Detailed steps of the inclusion and exclusion criteria of the study are shown in Fig 1. Briefly, a total of 703 studies were retrieved through five electronic database searches. Thirty-eight additional studies were identified from the references of included studies after removing the duplicates. On the basis of extraneous titles and abstracts, 526 studies were excluded. Several studies were excluded because they were reviews, non-human experiments; did not present the corresponding outcomes and lack usable data. Ultimately, the final meta-analysis was conducted after complete evaluation of 17 studies that met the inclusion criteria [10-14, 20-30].The basic characteristics and quality score of the included studies were presented in Table 1. The quality score of studies was proved to be acceptable (Table 2).

Fig. 1 Study search and selection flow diagram.

Relationship between serum IGF-1 and ischemic stroke risk

The relationship between serum IGF-1 and ischemic stroke patients was evaluated, and there was a high degree of heterogeneity among these included fifteen studies (I^2=99%). Thus, the random-effects model was utilized to pool MDs. The consequence manifested a significant correlation (MD = -45.32, 95% CI = [-63.70, -26.94], P < 0.00001, I^2=99%, random effect) (Fig. 2).

Fig. 2 Forest plot of the relationship between serum IGF-1 and ischemic stroke risk.

Relationship between serum IGF-1 and poststroke functional independence

Poststroke functional independence was identified by mRS at 90 days. The relationship between serum IGF-1 and poststroke functional independence was evaluated, and the random-effects model was utilized to pool MDs.

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The result manifested no significant difference (MD=27.52, 95% CI= [3.89, 51.14], P=0.02, I^2=96%, random effect) (Fig. 3).

Fig. 3 Forest plot of the relationship between serum IGF-1 and poststroke functional Independence.

Regression of heterogeneity sources

There is high heterogeneity in analyses so we use meta-regression to explore heterogeneity. From the analysis results, we can see that the sample size may be the primary source of heterogeneity (Table 3).

Table 3 Regression of heterogeneity sources

Item	Number of studies	95% CI	P value
Country	15	[-41.50, 78.27]	0.516
Sample size	15	[0.05, 0.28]	0.008
Male	14	[-4.42, 1.57]	0.320
The time from symptom onset to blood collection	14	[-18.71, 83.61]	0.192

Subgroup analysis of heterogeneity sources

In view of enormous heterogeneity, subgroup analysis of heterogeneity sources was performed to explore heterogeneity sources. Heterogeneity of each subgroup decreased modestly in varying degrees. Notably, the heterogeneity of the time from symptom onset to blood collection presented a significant decrease in parallel with other subgroup results (Table 4).

Table 4 Subgroup analysis of heterogeneity sources

Item	Subgroup	Number of studies	MD	95% CI	P value	Heterogeneity (I^2)
Country	Asian	12	-55.44	[-72.07, -38.82]	< 0.00001	97%
	Non-Asian	3	-2.92	[-32.42, 26.58]	0.85	99%
Sample size	≥100	6	-18.84	[-45.77, 8.08]	0.17	99%
	<100	9	-65.16	[-88.48, -41.83]	< 0.00001	97%
Male	≥60%	6	-65.89	[-107.51, -24.26]	0.02	98%
	<60%	8	-35.21	[-65.91, -4.52]	0.2	99%
The time from	≤48 hours	7	-30.04	[-40.05, -20.03]	< 0.00001	86%
	>48 hours	4	-52.40	[-120.79, 15.99]	0.13	99%

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symptom onset to blood collection	≤24 hours	5	-33.76	[-49.51, -18.01]	< 0.0001	89%

Abbreviations: MD, mean difference; CI, confidence interval.

Sensitivity analysis and adjusted relationship between serum IGF-1 and ischemic stroke

Sensitivity analysis was performed by omitting one study by turns, which showed that there was no significant influence of any individual study on the pooled MDs (Fig. 4). Moreover, in pursuit of low degree of heterogeneity and to uncover low degree of heterogeneity sources, the included studies were randomly merged. It was uncovered that heterogeneity reach to the minimum when four included studies were merged (MD=-30.32, 95% CI= [-36.52, -24.11], P< 0.00001, I^2=0%) (Fig.5 A) [10, 13, 21, 23]. In contrast, the other eleven included studies remained high degree of heterogeneity (MD=-50.34, 95% CI= [-72.82, -27.87], P<0.00001, I^2=99%) (Fig. 5 B) [11, 12, 14, 22, 24-30]. Taken together, high degree of heterogeneity seemed not to discredit the final merged result.

Fig. 4 Forest plot in the sensitivity analysis of serum IGF-1 and ischemic stroke risk.

Fig. 5 Forest plot of the adjusted relationship between serum IGF-1 and ischemic stroke risk in (A) four included studies and (B) eleven included studies.

Publication bias analysis

According to Begg's test (P = 0.092 > 0.05) and the funnel plot (Fig. 6), no potential publication bias among the 17 included studies was noted in this meta-analysis.

Fig. 6 Begg's funnel plot for publication bias test of the relationship between serum IGF-1 and ischemic stroke risk.

Discussion

The mechanism of serum IGF-1 in evaluating the risk and prognosis of ischemic stroke

This meta-analysis merged 17 studies comprising 2277 patients to evaluate the association of serum IGF-1 with the risk of incident ischemic stroke and the prognostic value of serum IGF-1 in ischemic stroke. The results demonstrated that serum IGF-1 could be utilized to evaluate the risk of incident ischemic stroke (MD = -45.32, 95% CI = [-63.70, -26.94], P < 0.00001, I^2=99%, random-effect) and the outcome after ischemic stroke (MD=27.52, 95% CI= [3.89, 51.14], P=0.02, I^2=96%, random-effect). The main mechanism is as follows: Serum IGF-1 mainly comes from the liver, a small part comes from the kidneys and peripherals [31]. In body fluids, IGF-1 has a high affinity with IGF binding proteins (IGFBPs). IGFBPs regulate the biological activity and

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bioavailability of IGF-1, and the intake of serum IGF-1 enters the cerebrospinal fluid through the blood-brain barrier (BBB) [32]. IGF-1 binds to the widely distributed receptor IGF-1 receptor (IGF-1R) in the brain to activate the two main signal pathways of PI3K/Akt and MAPK, thereby inhibiting neuronal apoptosis [33], promoting the survival of oligodendrocytes [34], and protecting the hippocampus [35].

Discussion on the causes of heterogeneity

However, there was inevitable high degree of heterogeneity among the included studies in this meta-analysis. In order to make the conclusion more persuasive and scientific, subgroup analysis was performed to explore the heterogeneity sources. As the results suggested, the time from symptom onset to blood collection appeared to be part of the source of heterogeneity, while heterogeneity decreased modestly in varying degrees. A study reported that stroke severity was related to decreased IGF-1 in rat as well as human, which meant that different stroke severity of recruited patients may also cause heterogeneity [11, 36]. Besides, insufficient follow-up duration may be the heterogeneity sources. Due to the inability to obtain the original data, the heterogeneity from the time from symptom onset to blood collection, follow-up duration and stroke severity of recruited patients cannot be explored further. By scrutinizing the included studies, it was found that ischemic stroke patients who were recruited were subjected to the World Health Organization criteria in some included studies [10, 12, 13], whereas others included studies were not subjected to the criteria. In addition to different patient recruitment criteria, the nutritional status of the patient may cause heterogeneity, which can be indirectly reflected by BMI. Many ischemic stroke patients are followed by accelerated catabolism and decreased food intake, which results in the body muscle mass and BMI decrease. The current study cannot rule out the possible effect of malnutrition on serum IGF-1. Moreover, forest plots in the sensitivity analysis (Fig. 4) and of the adjusted relationship between serum IGF-1 and ischemic stroke risk (Fig. 5) both manifest that the exclusion of any included study and a high degree of heterogeneity do not affect the final combined result.

The relationship between growth hormone and IGF-1 in cerebrovascular diseases

Growth hormone (GH) exerts much of biological activity by promoting the production of IGF-1, and IGF-1 can also mediate the effects of GH in turn. The majority of circulating serum IGF-1 are completed with IGFBPs and acid labile subunits, which prevent IGF-1 from lighting to the insulin receptor or modulate IGF-1 binding to the IGF-1R [37]. It is universally acknowledged that the bioavailability of IGF-1 is regulated by IGFBPs and only unbound or free IGF-1 possesses biological activity. As the body ageing process, IGF-1 decreases sharply. Some human studies demonstrate the effects of circulating serum IGF-1 in cerebrovascular events may be induced by metabolic syndrome [38, 39]. Moreover, patients suffering from growth hormone deficiency, as a result, may have lower circulating IGF-1 levels, which triggered considerably higher risk of cerebrovascular diseases [40]. Notably, it was reversible with growth hormone replacement [41].

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Insufficiency of previous related research

Previous studies assessing the association of IGF-1 with the risk of incident ischemic stroke and the prognostic value of serum IGF-1 in ischemic stroke reported controversial findings. A study performed by Dong et al. (2014) and colleagues suggested that lower IGF-I levels are significantly related to risk of stroke and stroke severity, but IGFBPs levels were not obtained in that biologically active IGF-I was not fully represented [11]. Johnson et al. (2005) [12] found inverse associations between levels of IGF-I and IGFBP-3 and the risk of ischemic stroke. However, another study showing no association with incident ischemic stroke [15]. Meanwhile, Aberg et al. (2011) [14] suggested that serum IGF-1 levels were closely related to improvements in the mRS and Scandinavian Stroke Scale tested at 3 and 24 months. However, the widespread variation of sampling may have to influence conclusions. Mattlage et al. (2016) [42] reported the potential importance of the change in IGF-1 and IGF-1 ratio to IGFBP-3 during the first week of stroke in understanding recovery. The limitation of the study was that serum IGF-1 levels were not supplemented with the levels of IGF-1 in the cerebral spinal fluid (CSF). Because in ischemic stroke, IGF-1 transfers from the blood to CSF. Nevertheless, another study showed inconsistent conclusion that higher serum IGF-1 levels were modestly associated with better outcome after ischemic stroke [20].

Strengths and limitations

To our knowledge, this meta-analysis is the first one to investigate relationship between serum IGF-1 levels and ischemic stroke. Shaheen et al. (2018) [43] have found that serum IGF-1 was an independent risk factor for ischemic stroke in Egyptian population which is consisitent with our study. This research has geographic limitations so this meta-analysis included both English and Chinese studies so that publication bias was avoided in a great deal. Besides, the eligibility criteria that study populations were not less than 25 cases and not more than 1000 cases to prevent large samples from affecting the overall results. Aberg et al. (2019) [44] have proved that levels of serum IGF-1 are associated with near 3-month outcomes. However, the study only included 354 patients so that the sample size is too small. Otherwise, a comprehensive database search was performed in this meta-anlysis, and as a result, 17 studies comprising a total of 2277 patients were included. Despite the shining points of this meta-analysis, its limitations and shortcomings should be emphasized. A high degree of heterogeneity may cause the unreliability of the results. Meanwhile, Begg's test of publication bias was not performed on the relationship between serum IGF-1 and poststroke functional independence due to the insufficient included studies. Ultimately, a data-feasible study [15] was excluded because of oversize cases, ambiguous patient recruitment criteria and no mention to stroke confirmed by imaging examination.

The clinical significance and future prospects of the study

Since most of the serum IGF-1 binds to IGFBPs, the current measured serum IGF level cannot represent the actual biological activity of igf in circulation. It is hoped that the future meta-analysis will use the data obtained

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by the IGF-1 specific tyrosine kinase receptor activation analysis method (KIRA) [32] as much as possible. This method indirectly measures the actual biological activity by accurately assessing the activity of the official receptor. This meta-analysis significantly analyzed that higher IGF-1 correlated with less risk of ischemic stroke. It provides a basis for in-depth clinical research on the relationship between the level of IGF-1 and neurological score [30] of stroke patients, post-stroke depression [45] dementi [46] and so on. In addition, it is also of great significance to the prevention of stroke (identification of high-risk groups and chemoprevention), early diagnosis, clinical treatment and prognosis.

Conclusions

This meta-analysis analyzed all the available interrelated information in the published studies and manifested that higher serum IGF-1 was significantly correlated with less risk of ischemic stroke. Concerning to the relationship between serum IGF-1 and outcome after ischemic stroke, it may need more studies to confirm the prognostic value of serum IGF-1 levels.

Competing interests

None declared.

Ethics Approval

None declared

Additional file

Additional file 1: PRISMA checklist. (DOC)

Authors' contributions

YNL* and WNY* are co-first authors. YNL and WNY designed the study. WNY, YSZ and JHL performed the literature search. WNY, YNL and LCZ selected the studies for inclusion in the meta-analysis. SMC and YNL extracted the data. WNY performed the statistical analysis. YNL, WNY and JHL wrote the manuscript. LH and YYZ revised the manuscript. YYZ provided the funding. All authors read and approved the final manuscript.

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Availability of data and materials

All data analyzed during this study are included in this article.

Ethics approval and consent to participate

Not applicable.

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Consent for publication

Not applicable.

Competing interests for Reviewer

None declared

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Table 1. Basic characteristics of included studies in the meta-analysis

Study	Study period	Country	The time from symptom onset to blood collection	Age (y) [a or b]	Male (%)	BMI (kg/m^2) [a or b]	Follow-up duration	IGF-I (ng/mL)			
								Risk (Mean, SD or IQR, Total)		mRS (Mean, SD, Total)	
								Case	Control	mRS 0-2	mRS 3-6
Fu et al. (2001) [22]	Not Stated	China	NA	57-74	55.00	NA	NA	205.7, 125.8, 20	472.6, 317.9, 20	NA	NA
Sun et al. (2001) [27]	1998.9-1999.10	China	≤24 hours	67	70.00	26.45±2.51	NA	91, 51 ,30	175, 49 ,30	NA	NA
Liu et al. (2002) [25]	1999.2-2002.3	China	3 days	63.85±8.74	63.33	27±5	NA	94, 27, 60	185, 30, 30	NA	NA
Denti et al. (2004) [10]	1998.1-2000.6	Italy	≤24 hours	83±7	34.12	25±5	6 months	69, 45, 85	102, 67, 88	NA	NA
Zhang et al. (2004) [29]	2000.12-2001.6	China	NA	64±11	66.67	NA	NA	99.48, 33.76, 57	179.9, 35.47, 26	NA	NA
Johnsen et al. (2005) [12]	1993.12-1997.5	Denmark	NA	50-64	61.00	26.5±4.0	NA	107.1, 33.3, 254	112.9, 33.2, 254	NA	NA

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Geng et al. (2006) [23]	2003.6-2004.1	China	≤48 hours	65.08± 10.93	70.27	NA	NA	153.22, 64.56, 56	196.1, 33.43, 10	NA	NA
Peng et al. (2009) [26]	2006.12-2008.6	China	≤48 hours	NA	NA	NA	NA	10.4, 1.6, 22	28.2, 7.8, 30	NA	NA
Chen et al. (2010) [21]	2009.1-2009.12	China	≤24 hours	65.64±7.59	55	NA	NA	73.5, 28 ,40	103.3, 5.8, 20	NA	NA
Aberg et al. (2011) [14]	1998.8-2003.12	Western Sweden	4 days	55.5±0.86	56.65	26.2±0.43	3 months	172, 3, 388	145, 9, 40	NA	NA
Dong et al. (2014) [11]	2011.8-2013.8	China	≤24 hours	66 (57–77)	58.75	23.9(22.6–26.8)	NA	129, 44, 221	140, 34, 200	NA	NA
Tang et al. (2014) [13]	2012.8-2013.8	China	≤24 hours	72 (62–83)	53.57	25.3(21.8–28.4)	3 months	127, 50, 168	155, 38, 100	114, 29, 113	146, 43, 55
Zhang et al. (2016) [30]	2013.2-2015.9	China	NA	65.2±8.6	56.00	26.28+1.59	3 months	87.92, 25.31, 50	152.21, 29.32, 50	NA	NA
Li et al. (2016) [24]	2014.5-2015.6	China	3 days	51.9±9.6	56.41	23.9±6.1	NA	98.66, 20.91, 39	153.01, 26.35, 43	NA	NA
Armbrust et al. (2017)	NA	NA	≤24 hours	72.9±9.5	55.1	26.8±4.4	3 months	NA	NA	131.7, 55.7, 315	114.2, 55.3, 89

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[19]											
Tao et al. (2018) [28]	2016.1-2017.3	China	3 days	60.30±9.75	66.67	NA	3 months	106.54, 48.84, 60	199.2, 53.15, 30	145.81, 49.89, 57	85.56, 32.16, 48
Aberg et al. (2019) [20]	1998-2003	Western Sweden	≤24 hours	55.3±11	65.12	NA	7 years	NA	NA	151.5, 6.6, 256	148.3, 12.3, 67

[a] Median (IQR); [b] Mean (SD).

Abbreviations: BMI, body mass index; NA, not available; IQR, interquartile range; SD, standard deviation.

Table 2 Newcastle-Ottawa Scale for quality assessment

Study	Exposed cohort	Non-exposed cohort	Ascertainment of exposure	Outcome of interest	Control for factor	Assessment of outcome	Follow-up long enough	Adequacy of follow-up	Total score
Fu et al. (2001) [22]	*	*	*		**	*			6
Sun et al. (2001) [27]	*	*	*		**	*			6
Liu et al. (2002) [25]	*	*	*		**	*			6
Denti et al. (2004) [10]	*	*	*		**	*	*	*	8

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Zhang et al. (2004) [29]	*	*	*		**	*			6
Johnsen et al. (2005) [12]	*	*	*		**	*			6
Geng et al. (2006) [23]	*	*	*		**	*	*		7
Peng et al. (2009) [26]	*	*	*		**	*			6
Chen et al. (2010) [21]	*	*	*		**	*	*		7
Aberg et al. (2011) [14]	*	*	*		**	*	*	*	8
Dong et al. (2014) [11]	*	*	*		**	*			6
Tang et al. (2014) [13]	*	*	*		**	*	*	*	8
Zhang et al. (2016) [30]	*	*	*		**	*	*	*	8
Li et al. (2016)	*	*	*		**	*			6

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[24]									
Armbrust et al. (2017) [19]	*	*	*		**	*	*	*	8
Tao et al. (2018) [28]	*	*	*		**	*	*	*	8
Aberg et al. (2019) [20]	*	*	*	*	**	*	*	*	9

*The study scored one point in the project; **The study scored two points in the project.

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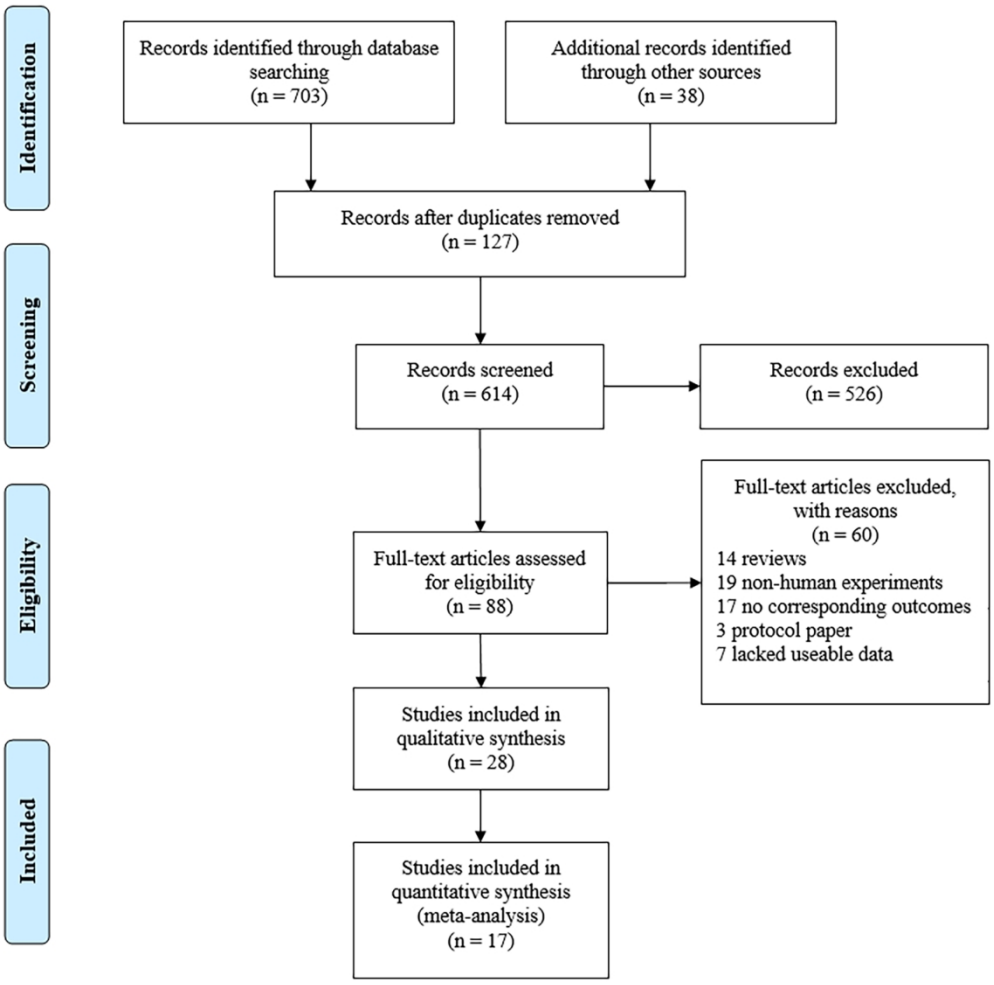


Figure 1

47x46mm (1000 x 1000 DPI)

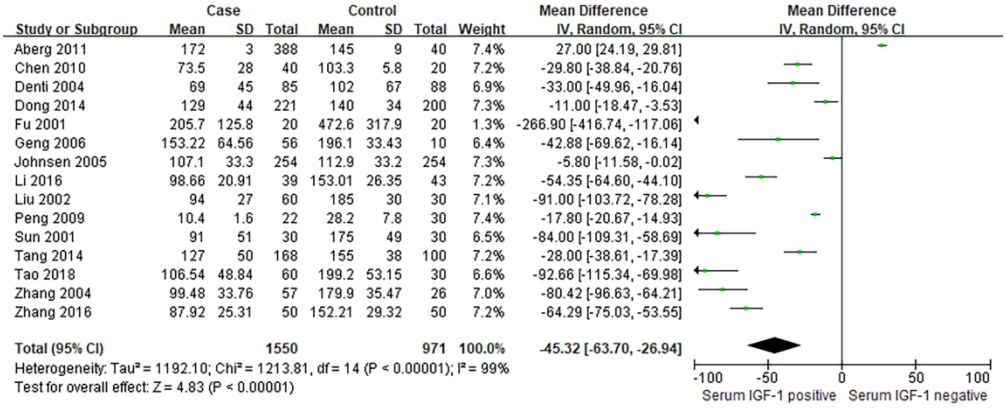


Figure 2

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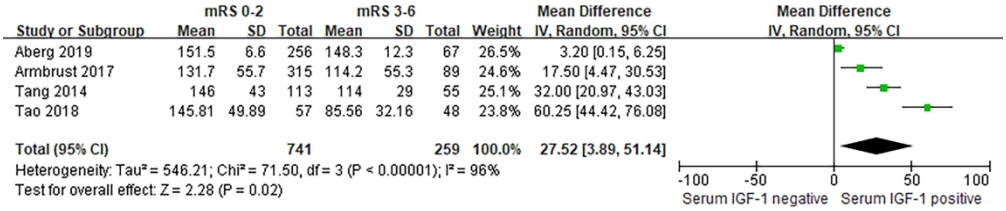


Figure 3

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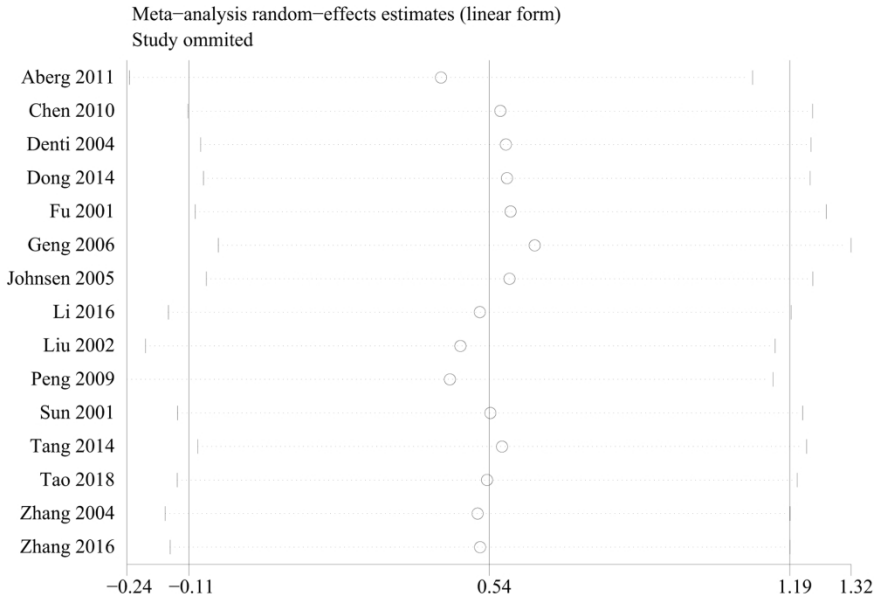


Figure 4

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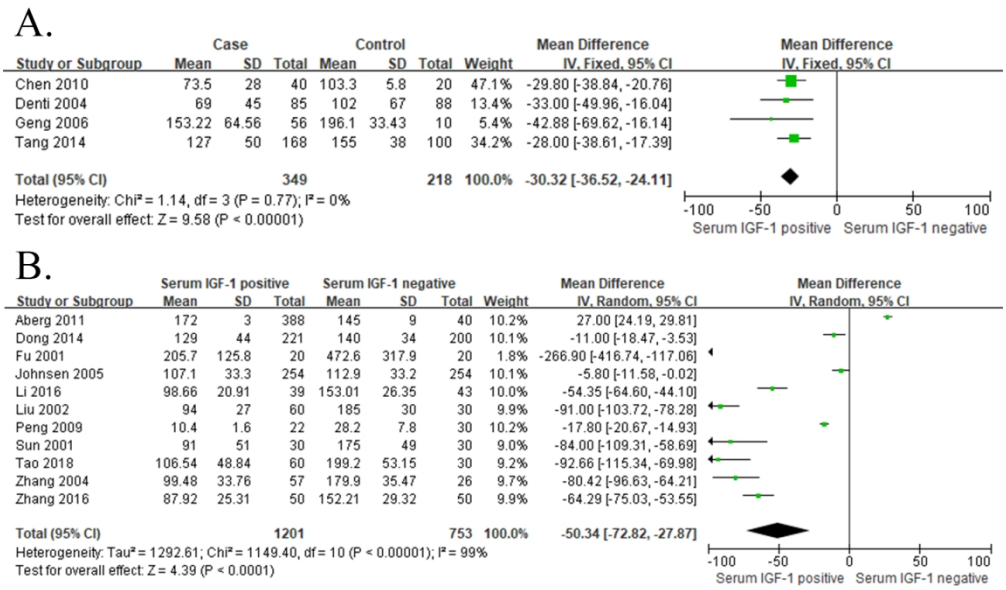


Figure 5

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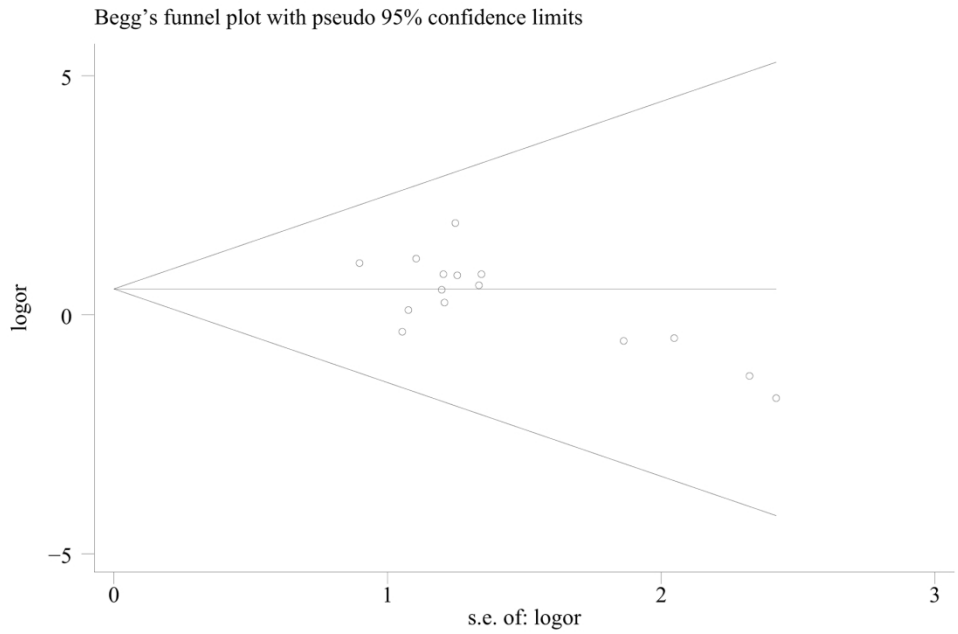

Begg's funnel plot with pseudo 95% confidence limits
5
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Figure 6

Search strategy

This is a systematic review and meta-analysis about the relationship between serum insulin-like growth factor 1 levels and ischemic stroke. To ensure the rationality and extensiveness of data search, detailed database retrieval strategies for retrieval process will presented as followings.

Step1: Clarify retrieval problems and requirements

We transform the problem into the PICO model.

Step2: Clear search source

PubMed, Scopus, Embase, China National Knowledge Infrastructure (CNKI) and Web of Science were searched for studies published or presented on or before December 29, 2020. For maximum search, publications are not limited to English.

Step3: Determine search terms

According to the study content, we have identified "IGF-1" and "stroke" as search terms. To achieve maximum sensitivity of the search strategy and to include more extensive studies, Medical Subject Headings (MeSH) terms and the keywords that were retrieved in the fifield of Title/Abstract were used in combination.

After searching, we determined the subject words and free words as follows:

Subject words: Insulin-Like Growth Factor I

Free words: Insulin-Like Somatomedin Peptide I; Insulin Like Somatomedin Peptide I; Somatomedin C; IGF-I-SmC; IGF-1; IGF-I; Insulin Like Growth Factor I.

Subject words: Stroke

Free words: Strokes; Cerebrovascular Accident; Cerebrovascular Accidents; CVA (Cerebrovascular Accident); CVAs (Cerebrovascular Accident); Cerebrovascular Apoplexy; Apoplexy, Cerebrovascular; Vascular Accident, Brain; Brain Vascular Accident; Brain Vascular Accidents; Vascular Accidents, Brain; Cerebrovascular Stroke; Cerebrovascular Strokes; Stroke, Cerebrovascular; Strokes, Cerebrovascular; Apoplexy; Cerebral Stroke; Cerebral Strokes; Stroke, Cerebral; Strokes, Cerebral; Stroke, Acute; Acute Stroke; Acute Strokes; Strokes, Acute; Cerebrovascular Accident, Acute; Acute Cerebrovascular Accident; Acute Cerebrovascular Accidents; Cerebrovascular Accidents, Acute.

" IGF-1" was selected in the MeSH term and "Insulin-Like Somatomedin Peptide I", "Insulin Like Somatomedin Peptide I", "Somatomedin C", "IGF-I-SmC", "IGF-I", "Insulin Like Growth Factor I"

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were retrieved in the fifield of Title/Abstract. All the above were connected by “OR”. “Stroke” was selected in the MeSH term. Ultimately, “AND” was used to connect the two parts.

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	Title page, page 1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	Abstract, page 2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	Introduction, page 2, 3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	Introduction, page 2, 3
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	Study protocol, page 3
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	Eligibility criteria, page 3
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	Literature search strategy, page 3
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Literature search strategy, page 3
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	Literature search strategy, page 3
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	Data extraction and quality assessment, page 3
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	Data extraction and quality assessment, page 3
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	Data extraction and quality assessment, page 3
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	Statistical analysis, page 4
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	Statistical analysis, page 4

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	Statistical analysis, page 4
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	Statistical analysis, page 4
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	Study selection and characteristics, Figure 1, page 4
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	Study selection and characteristics, Table 1, page 4
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	Publication bias analysis and sensitivity analysis, Fig. 6, page 5
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	Fig. 2-5
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	Results, page 4, 5
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	Publication bias analysis and sensitivity analysis, Fig. 6, page 5
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Results, Table 3, Fig 4,5, page 4, 5
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	Discussion, page 6
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	Discussion, page 6, 7
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	Discussion, page 5-7
FUNDING			

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PRISMA 2009 Checklist

Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	Funding, page 7

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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Primary Subject Heading:	Neurology
Secondary Subject Heading:	Neurology
Keywords:	Stroke < NEUROLOGY, Biochemistry < NATURAL SCIENCE DISCIPLINES, Adult neurology < NEUROLOGY

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The relationship between serum insulin-like growth factor 1 levels and ischemic stroke: a systematic review and meta-analysis

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Abbreviations

IGF-1: insulin-like growth factor 1; MD: mean difference; CI: confidence interval; IGFs: Insulin-like growth factors; NO: nitric oxide; CNKI: China National Knowledge Infrastructure; CT: computed tomography; MRI: magnetic resonance imaging; BMI: body mass index; NIHSS: National Institutes of Health Stroke Scale; mRS: modified Rankin Scale; NOS: Newcastle-Ottawa Scale; GH: Growth hormone; IGFBPs: insulin-like growth factor binding proteins; IGF-1R: insulin-like growth factor 1 receptor; CSF: cerebral spinal fluid; PRISMA: preferred reporting items for systematic reviews and meta-analyses

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Abstract

Objective: To assess the association of insulin-like growth factor 1 (IGF-1) with the risk of incident ischemic stroke and outcome after ischemic stroke.

Design: A systematic review of primary studies.

Setting: Hospitals in Western Sweden, Italy, China and Denmark.

Methods: A search was carried out in eligible studies in electronic databases (PubMed, Scopus, Embase, China National Knowledge Infrastructure and Web of Science) updated to December 29, 2020. The relevant data were extracted in order to conduct the meta-analysis. Review Manager version 5.2 was utilized to pool data and calculate the mean difference (MD) and its 95% confidence interval (CI). Heterogeneity, subgroup analysis, sensitivity analysis and publication bias were also performed in this meta-analysis.

Results: A total of 2277 patients were included in 17 studies. This meta-analysis indicated that higher serum IGF-1 levels were significantly correlated with less risk of ischemic stroke (MD=-45.32 95% CI = [-63.70, -26.94], P < 0.00001, I^2=99%) and better improvement of outcome after ischemic stroke (MD=27.52, 95% CI= [3.89, 51.14], P=0.02, I^2=96%). According to subgroup analysis, heterogeneity comes from country, sample size, male and the time from symptom onset to blood collection. Sensitivity analysis showed that there was no significant influence of any individual study on the pooled mean difference. The effect of high heterogeneity on result credibility was eliminated when four included studies were merged (MD=-30.32, 95% CI= [-36.52, -24.11], P< 0.00001, I^2=0%). Moreover, no potential publication bias was discovered in this meta-analysis.

Conclusion: Higher serum IGF-1 was significantly correlated with a lower risk of ischemic stroke. In view of the high degree of heterogeneity, it may need more studies to confirm the prognostic value of serum IGF-1 levels in ischemic stroke and explore the the sources of heterogeneity.

Keywords: insulin-like growth factor-1, outcome, ischemic stroke, risk.

Strengths and limitations of this study

- This is a meta-analysis of 17 published studies comprising a total of 2277 patients in this study.
- The included studies were heterogeneous in the time from symptom onset to blood collection and follow-up duration at the end of the study.
- Study populations were limited to not less than 25 cases and no more than 1000 cases to prevent large samples from affecting the overall results.
- A thorough systematic review and meta-analysis provide confidence in the findings.
- Begg's test of publication bias was not performed on the relationship between serum IGF-1 and poststroke functional independence due to insufficient included studies.

Introduction

Insulin-like growth factors (IGFs), confirmed as peptide hormones, share similar structural homology with insulin. Insulin-like growth factor 1 (IGF-1) protects cells from apoptosis and functions as a vital regulator in cell survival [1, 2]. A growing body of evidence has established that IGF-1 influenced vascular function in many

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different ways, possessing anti-inflammatory properties and inducing vasodilation by promoting nitric oxide (NO) production [3, 4]. Recently, substantial evidence has accrued to demonstrate that IGF-1 plays a significant role in many cerebral diseases, for example, depression in ischemic stroke, cerebrovascular disease and Alzheimer's disease [5-7].

Stroke is a leading cause of death and disability worldwide, especially in the current aging society [8]. Animal studies and observational studies on humans have been trying to establish an association between IGF-1 and the risk of ischemic stroke. However, the association of IGF-1 with the risk of incident ischemic stroke still remains unclear. Seven previous studies are exploring the association manifested inconsistent conclusions. On the one hand, five studies showed that IGF-1 possessed a protective effect against ischemic stroke [9-13]. On the other hand, no association between incident ischemic stroke and IGF-1 was uncovered in two studies [14, 15]. Moreover, IGF-1 is widely disseminated and highly expressed in the nervous system [16], which exerts neuroprotective effects and accelerates proliferation of neurons. According to some studies, circulating serum IGF-1 is correlated with improvement of functional outcome after ischemic stroke and might predict outcome after ischemic stroke [13, 14, 17-19]. Actually, the outcome after ischemic stroke relies on complex interactions of multiple factors, and it still needs more studies to disclose whether the association is causative.

This meta-analysis was conducted in order to evaluate the association of IGF-1 with the risk of incident ischemic stroke and the outcome after ischemic stroke.

Methods

Patient and Public Involvement

This article didn't involve patients and or public.

Study protocol

This meta-analysis was carried out in accordance with the preferred reporting items for systematic reviews and meta-analyses (PRISMA) statement (Additional file 1).

Literature search strategy

PubMed, Scopus, Embase, China National Knowledge Infrastructure (CNKI) and Web of Science were searched for studies published or presented on or before December 29, 2020. Publications are not restricted to English. To achieve maximum sensitivity in the search strategy and to include more extensive studies, Medical Subject Headings (MeSH) terms and the keywords that were retrieved in the field of Title/Abstract were used in combination. "IGF-1" was selected in the MeSH term and "Insulin-Like Somatomedin Peptide I", "Insulin Like Somatomedin Peptide I", "Somatomedin C", "IGF-I-SmC", "IGF-I" and "Insulin Like Growth Factor I" were retrieved in the field of Title/Abstract. All the above are connected by "OR". "Stroke" was selected in the MeSH term. Ultimately, "AND" was utilized to connect the two parts. All relevant articles were found first, and then

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irrelevant articles were manually excluded through title and abstract screening and full-text assessment. To avoid selection bias，all work was carried out independently by two authors.

Eligibility criteria

Studies were included if they fulfilled the following criteria:

- Studies were confined to those involving human subjects.
- There were no restrictions on the information of language, year of the study, patient age, imaging criteria, body mass index (BMI) and the National Institutes of Health Stroke Scale (NIHSS) score.
- Study populations had no fewer than 25 cases and no more than 1000 cases.
- Study had a definite inclusion criterion for acute ischemic stroke.
- Patients who suffered from stroke were confirmed by computed tomography (CT) or magnetic resonance imaging (MRI) before treatment.
- Studies reported the complete outcomes.
- Study used a prospective or retrospective cohort design, or nested case–control design.
- If duplicate studies with an accumulating number of patients or something else were published, only the most complete studies were included in the analysis performed in this study.

Data extraction and quality assessment

All data were extracted independently from the tables and figures of included studies by two authors in an unblinded fashion or adjudicated by the third author in case of disagreement. Moreover, another author took responsibility for the accuracy and reliability of all of the extracted data. The modified Rankin Scale (mRS) is a clinical poststroke functional independence assessment tool.. For each study, the study characteristics (author, publication year, study period, country,), main characteristics of the patients (age gender and BMI) and outcomes (the number of ischemic stroke patients, the number of people in control, serum IGF-1 and mRS at 90 days) were extracted independently by two authors. The quality of the included studies was assessed by two other authors using the modified Newcastle-Ottawa Scale (NOS), while 6 points or more was considered to be of high quality.

Outcome measure

The Mean, SD, Total of serum IGF-1 (ng/mL) in case and control groups is a main indicator of stroke risk assessment. The mRS is an evaluation tool to assess the clinical poststroke functional independence (detailed grades of mRS: 0 - No symptoms; 1 - No significant disability; 2 - Slight disability; 3 -Moderate disability; 4 - Moderately severe disability; 5 - Severe disability; 6 - Dead). The prespecified primary outcome measures were the mRS score of 0 - 2 and the mRS score 3 - 6.

Statistical analyses

Statistical analysis was implemented using Review Manager version 5.2 (The Nordic Cochrane Centre, the

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Cochrane Collaboration, Copenhagen, Denmark). The combined effect of mean difference (MD) and 95% confidence interval (CI) was defined by a random-effect model or a fixed-effect model. The heterogeneity among included studies was tested using the inconsistency index (I^2). A random-effect model was used when there was a high level of heterogeneity (I^2 values > 50%). A P value <0.05 was considered statistically significant. For indicators with higher heterogeneity, sensitivity analysis and subgroup analyses were carried out to identify the reliability of the meta-analysis and explore the sources of heterogeneity in the Stata 12.0 software. Publication bias was evaluated qualitatively by visual observation of funnel graphs and was assessed quantitatively by the Begg's test (Stata 12.0), which manifests a publication bias with P <0.05. The number, mean difference (MD), standard deviation (SD) of the experimental group and control group of the included studies and country, sample size, male, the time from symptom onset to blood collection of 17 included studies were imported into Stata 12.0 software.

Results

Study selection and characteristics

Detailed steps of the inclusion and exclusion criteria of the study are shown in Fig 1. Briefly, a total of 703 studies were retrieved through five electronic database searches. Thirty-eight additional studies were identified from the references of included studies after removing the duplicates. On the basis of extraneous titles and abstracts, 526 studies were excluded. Several studies were excluded because they were reviews, non-human experiments; did not present the corresponding outcomes and lack usable data. Ultimately, the final meta-analysis was conducted after a complete evaluation of 17 studies that met the inclusion criteria [10-14, 20-30].The basic characteristics and quality score of the included studies are presented in Table 1. The quality score of studies was proven to be acceptable (Table 2).

Fig. 1 Study search and selection flow diagram.

Relationship between serum IGF-1 and ischemic stroke risk

The relationship between serum IGF-1 and ischemic stroke patients was evaluated, and there was a high degree of heterogeneity among these included fifteen studies (I^2=99%). Thus, the random-effects model was utilized to pool MDs. The consequence manifested a significant correlation (MD = -45.32, 95% CI = [-63.70, -26.94], P < 0.00001, I^2=99%, random effect) (Fig. 2).

Fig. 2 Forest plot of the relationship between serum IGF-1 and ischemic stroke risk.

Relationship between serum IGF-1 and poststroke functional independence

Poststroke functional independence was identified by mRS at 90 days. The relationship between serum IGF-1 and poststroke functional independence was evaluated, and the random-effects model was utilized to pool MDs. The result showed no significant difference (MD=27.52, 95% CI= [3.89, 51.14], P=0.02, I^2=96%, random effect)

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(Fig. 3).

Fig. 3 Forest plot of the relationship between serum IGF-1 and poststroke functional Independence.

Regression of heterogeneity sources

There is high heterogeneity in analyses so we use meta-regression to explore heterogeneity. From the analysis results, we can see that the sample size may be the primary source of heterogeneity (Table 3).

Table 3 Regression of heterogeneity sources

Item	Number of studies	95% CI	P value
Country	15	[-41.50, 78.27]	0.516
Sample size	15	[0.05, 0.28]	0.008
Male	14	[-4.42, 1.57]	0.320
The time from symptom onset to blood collection	14	[-18.71, 83.61]	0.192

Subgroup analysis of heterogeneity sources

In view of enormous heterogeneity, subgroup analysis of heterogeneity sources was performed to explore heterogeneity sources. The heterogeneity of each subgroup decreased modestly to varying degrees. Notably, the heterogeneity of the time from symptom onset to blood collection presented a significant decrease in parallel with other subgroup results (Table 4).

Table 4 Subgroup analysis of heterogeneity sources

Item	Subgroup	Number of studies	MD	95% CI	P value	Heterogeneity (I^2)
Country	Asian	12	-55.44	[-72.07, -38.82]	< 0.00001	97%
	Non-Asian	3	-2.92	[-32.42, 26.58]	0.85	99%
Sample size	≥100	6	-18.84	[-45.77, 8.08]	0.17	99%
	<100	9	-65.16	[-88.48, -41.83]	< 0.00001	97%
Male	≥60%	6	-65.89	[-107.51, -24.26]	0.02	98%
	<60%	8	-35.21	[-65.91, -4.52]	0.2	99%
The time from symptom	≤48 hours	7	-30.04	[-40.05, -20.03]	< 0.00001	86%
	>48 hours	4	-52.40	[-120.79, 15.99]	0.13	99%
	≤24 hours	5	-33.76	[-49.51, -18.01]	< 0.0001	89%

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onset to blood collection						

Abbreviations: MD, mean difference; CI, confidence interval.

Sensitivity analysis and adjusted relationship between serum IGF-1 and ischemic stroke

Sensitivity analysis was performed by omitting one study by turn, which showed that there was no significant influence of any individual study on the pooled MDs (Fig. 4). Moreover, in pursuit of a low degree of heterogeneity and to uncover low degree of heterogeneity sources, the included studies were randomly merged. It was discovered that heterogeneity reached its minimum when four included studies were merged (MD=-30.32, 95% CI= [-36.52, -24.11], P< 0.00001, I^2=0%) (Fig.5 A) [10, 13, 21, 23]. The other eleven included studies, in the other hand, remained high degree of heterogeneity (MD=-50.34, 95% CI= [-72.82, -27.87], P<0.00001, I^2=99%) (Fig. 5 B) [11, 12, 14, 22, 24-30]. Taken together, high degree of heterogeneity seemed not to discredit the final merged result.

Fig. 4 Forest plot in the sensitivity analysis of serum IGF-1 and ischemic stroke risk.

Fig. 5 Forest plot of the adjusted relationship between serum IGF-1 and ischemic stroke risk in (A) four included studies and (B) eleven included studies.

Publication bias analysis

According to Begg's test (P = 0.092 > 0.05) and the funnel plot (Fig. 6), no potential publication bias among the 17 included studies was noted in this meta-analysis.

Fig. 6 Begg's funnel plot for publication bias test of the relationship between serum IGF-1 and ischemic stroke risk.

Discussion

The mechanism of serum IGF-1 in evaluating the risk and prognosis of ischemic stroke

This meta-analysis merged 17 studies comprising 2277 patients to evaluate the association of serum IGF-1 with the risk of incident ischemic stroke and the prognostic value of serum IGF-1 in ischemic stroke. The results demonstrated that serum IGF-1 could be utilized to evaluate the risk of incident ischemic stroke (MD = -45.32, 95% CI = [-63.70, -26.94], P < 0.00001, I^2=99%, random-effect) and the outcome after ischemic stroke (MD=27.52, 95% CI= [3.89, 51.14], P=0.02, I^2=96%, random-effect). The main mechanism is as follows: Serum IGF-1 mainly comes from the liver, a small part comes from the kidneys and peripherals [31]. In body fluids, IGF-1 has a high affinity with IGF binding proteins (IGFBPs). IGFBPs regulate the biological activity and bioavailability of IGF-1, and the intake of serum IGF-1 enters the cerebrospinal fluid through the blood-brain

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barrier (BBB) [32]. IGF-1 binds to the widely distributed receptor IGF-1 receptor (IGF-1R) in the brain to activate the two main signal pathways of PI3K/Akt and MAPK, thereby inhibiting neuronal apoptosis [33], promoting the survival of oligodendrocytes [34], and protecting the hippocampus [35].

Discussion on the causes of heterogeneity

However, there was an inevitable high degree of heterogeneity among the included studies in this meta-analysis. In order to make the conclusion more persuasive and scientific, subgroup analysis was performed to explore the heterogeneity sources. As the results suggested, the time from symptom onset to blood collection appeared to be part of the source of heterogeneity, while heterogeneity decreased modestly to varying degrees. A study reported that stroke severity was related to decreased IGF-1 in rats as well as humans, which meant that the different stroke severity of recruited patients may also cause heterogeneity [11, 36]. Besides, insufficient follow-up duration may be the heterogeneity source. Due to the inability to obtain the original data, the heterogeneity from the time of symptom onset to blood collection, follow-up duration and stroke severity of recruited patients cannot be explored further. By scrutinizing the included studies, it was found that ischemic stroke patients who were recruited were subjected to the World Health Organization criteria in some included studies [10, 12, 13], whereas others included studies were not subjected to the criteria. In addition to different patient recruitment criteria, the nutritional status of the patient may cause heterogeneity, which can be indirectly reflected by BMI. Many ischemic stroke patients are followed by accelerated catabolism and decreased food intake, which results in a decrease in body muscle mass and BMI. The current study cannot rule out the possible effect of malnutrition on serum IGF-1. Moreover, forest plots in the sensitivity analysis (Fig. 4) and of the adjusted relationship between serum IGF-1 and ischemic stroke risk (Fig. 5) both demonstrate that the exclusion of any included study and a high degree of heterogeneity do not affect the final combined result.

The relationship between growth hormone and IGF-1 in cerebrovascular diseases

Growth hormone (GH) exerts much of its biological activity by promoting the production of IGF-1, and IGF-1 can also mediate the effects of GH in turn. The majority of circulating serum IGF-1 is completed with IGFBPs and acid labile subunits, which prevent IGF-1 from binding to the insulin receptor or modulate IGF-1's binding to the IGF-1R [37]. It is universally acknowledged that the bioavailability of IGF-1 is regulated by IGFBPs and that only unbound or free IGF-1 possesses biological activity. As part of the body's aging process, IGF-1 decreases sharply. Some human studies demonstrate that the effects of circulating serum IGF-1 on cerebrovascular events may be induced by metabolic syndrome [38, 39]. Furthermore, patients with growth hormone deficiency may have lower circulating IGF-1 levels, which raises the risk of cerebrovascular diseases significantly [40]. Notably, it was reversible with growth hormone replacement [41].

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Insufficiency of previous related research

Previous studies assessing the association of IGF-1 with the risk of incident ischemic stroke and the prognostic value of serum IGF-1 in ischemic stroke reported controversial findings. A study performed by Dong et al. (2014) and colleagues suggested that lower IGF-I levels are significantly related to risk of stroke and stroke severity, but IGFBPs levels were not obtained in that biologically active IGF-I was not fully represented [11]. Johnson et al. (2005) [12] found inverse associations between levels of IGF-I and IGFBP-3 and the risk of ischemic stroke. However, another study showed no association with incident ischemic stroke [15]. Meanwhile, Aberg et al. (2011) [14] suggested that serum IGF-1 levels were closely related to improvements in the mRS and Scandinavian Stroke Scale tested at 3 and 24 months. However, the widespread variation of sampling may have influenced conclusions. Mattlage et al. (2016) [42] reported the potential importance of the change in IGF-1 and IGF-1 ratio to IGFBP-3 during the first week of stroke in understanding recovery. The limitation of the study was that serum IGF-1 levels were not supplemented with the levels of IGF-1 in the cerebral spinal fluid (CSF). Because in ischemic stroke, IGF-1 transfers from the blood to the CSF. Nevertheless, another study showed an inconsistent conclusion that higher serum IGF-1 levels were modestly associated with better outcomes after ischemic stroke [20].

Strengths and limitations

To our knowledge, this meta-analysis is the first one to investigate the relationship between serum IGF-1 levels and ischemic stroke. Shaheen et al. (2018) [43] have found that serum IGF-1 was an independent risk factor for ischemic stroke in the Egyptian population which is consistent with our study. This research has geographic limitations so this meta-analysis included both English and Chinese studies so that publication bias was avoided to a great extent. Besides, the eligibility criteria that study populations were not less than 25 cases and not more than 1000 cases to prevent large samples from affecting the overall results. Aberg et al. (2019) [44] have proved that levels of serum IGF-1 are associated with near 3-month outcomes. However, the study only included 354 patients, maening that the sample size is too small. Otherwise, a comprehensive database search was performed in this meta-anlysis, and as a result, 17 studies comprising a total of 2277 patients were included. Despite the shining points of this meta-analysis, its limitations and shortcomings should be emphasized. A high degree of heterogeneity may cause the unreliability of the results. Meanwhile, Begg's test of publication bias was not performed on the relationship between serum IGF-1 and poststroke functional independence due to insufficient included studies. Ultimately, a data-feasible study [15] was excluded because of oversize cases, ambiguous patient recruitment criteria and no mention of stroke confirmed by imaging examination.

The clinical significance and future prospects of the study

Because the majority of serum IGF-1 binds to IGFBPs, the current measured serum IGF level cannot accurately represent the biological activity of IGF in circulation. It is hoped that the future meta-analysis will use

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the data obtained by the IGF-1 specific tyrosine kinase receptor activation analysis method (KIRA) [32] as much as possible. This method indirectly measures the actual biological activity by accurately assessing the activity of the official receptor. This meta-analysis significantly analyzed that higher IGF-1 correlated with a lower risk of ischemic stroke. It provides a basis for in-depth clinical research on the relationship between the level of IGF-1 and the neurological score [30] of stroke patients, post-stroke depression [45] dementia [46] and so on. In addition, it is also of great significance to the prevention of stroke (identification of high-risk groups and chemoprevention), early diagnosis, clinical treatment and prognosis.

Conclusions

This meta-analysis analyzed all the available interrelated information in the published studies and found that higher serum IGF-1 was significantly correlated with a lower risk of ischemic stroke. Concerning the relationship between serum IGF-1 and outcome after ischemic stroke, it may need more studies to confirm the prognostic value of serum IGF-1 levels.

Competing interests

None declared.

Ethics Approval

None declared

Additional file

Additional file 1: PRISMA checklist. (DOC)

Authors' contributions

YNL* and WNY* are co-first authors. YNL and WNY designed the study. WNY, YSZ and JHL performed the literature search. WNY, YNL and LCZ selected the studies for inclusion in the meta-analysis. SMC and YNL extracted the data. WNY performed the statistical analysis. YNL, WNY and JHL wrote the manuscript. LH and YYZ revised the manuscript. YYZ provided the funding. All authors read and approved the final manuscript.

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Availability of data and materials

All data analyzed during this study are included in this article.

Ethics approval and consent to participate

Not applicable.

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Consent for publication

Not applicable.

Competing interests for Reviewer

None declared

Acknowledgments

Not applicable

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Table 1. Basic characteristics of included studies in the meta-analysis

Study	Study period	Country	The time from symptom onset to blood collection	Age (y) [a or b]	Male (%)	BMI (kg/m^2) [a or b]	Follow-up duration	IGF-I (ng/mL)			
								Risk (Mean, SD or IQR, Total)		mRS (Mean, SD, Total)	
								Case	Control	mRS 0-2	mRS 3-6
Fu et al. (2001) [22]	Not Stated	China	NA	57-74	55.00	NA	NA	205.7, 125.8, 20	472.6, 317.9, 20	NA	NA
Sun et al. (2001) [27]	1998.9-1999.10	China	≤24 hours	67	70.00	26.45±2.51	NA	91, 51 ,30	175, 49 ,30	NA	NA
Liu et al. (2002) [25]	1999.2-2002.3	China	3 days	63.85±8.74	63.33	27±5	NA	94, 27, 60	185, 30, 30	NA	NA
Denti et al. (2004) [10]	1998.1-2000.6	Italy	≤24 hours	83±7	34.12	25±5	6 months	69, 45, 85	102, 67, 88	NA	NA
Zhang et al. (2004) [29]	2000.12-2001.6	China	NA	64±11	66.67	NA	NA	99.48, 33.76, 57	179.9, 35.47, 26	NA	NA
Johnsen et	1993.12-1997.5	Denmark	NA	50-64	61.00	26.5±4.0	NA	107.1,	112.9,	NA	NA

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al. (2005) [12]								33.3, 254	33.2, 254		
Geng et al. (2006) [23]	2003.6-2004.1	China	≤48 hours	65.08± 10.93	70.27	NA	NA	153.22, 64.56, 56	196.1, 33.43, 10	NA	NA
Peng et al. (2009) [26]	2006.12-2008.6	China	≤48 hours	NA	NA	NA	NA	10.4, 1.6, 22	28.2, 7.8, 30	NA	NA
Chen et al. (2010) [21]	2009.1-2009.12	China	≤24 hours	65.64±7.59	55	NA	NA	73.5, 28 ,40	103.3, 5.8, 20	NA	NA
Aberg et al. (2011) [14]	1998.8-2003.12	Western Sweden	4 days	55.5±0.86	56.65	26.2±0.43	3 months	172, 3, 388	145, 9, 40	NA	NA
Dong et al. (2014) [11]	2011.8-2013.8	China	≤24 hours	66 (57–77)	58.75	23.9(22.6–26.8)	NA	129, 44, 221	140, 34, 200	NA	NA
Tang et al. (2014) [13]	2012.8-2013.8	China	≤24 hours	72 (62–83)	53.57	25.3(21.8–28.4)	3 months	127, 50, 168	155, 38, 100	114, 29, 113	146, 43, 55
Zhang et al. (2016) [30]	2013.2-2015.9	China	NA	65.2±8.6	56.00	26.28+1.59	3 months	87.92, 25.31, 50	152.21, 29.32, 50	NA	NA

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Li et al. (2016) [24]	2014.5-2015.6	China	3 days	51.9±9.6	56.41	23.9±6.1	NA	98.66, 20.91, 39	153.01, 26.35, 43	NA	NA
Armbrust et al. (2017) [19]	NA	NA	≤24 hours	72.9±9.5	55.1	26.8±4.4	3 months	NA	NA	131.7, 55.7, 315	114.2, 55.3, 89
Tao et al. (2018) [28]	2016.1-2017.3	China	3 days	60.30±9.75	66.67	NA	3 months	106.54, 48.84, 60	199.2, 53.15, 30	145.81, 49.89, 57	85.56, 32.16, 48
Aberg et al. (2019) [20]	1998-2003	Western Sweden	≤24 hours	55.3±11	65.12	NA	7 years	NA	NA	151.5, 6.6, 256	148.3, 12.3, 67

[a] Median (IQR); [b] Mean (SD).

Abbreviations: BMI, body mass index; NA, not available; IQR, interquartile range; SD, standard deviation.

Table 2 Newcastle-Ottawa Scale for quality assessment

Study	Exposed cohort	Non-exposed cohort	Ascertainment of exposure	Outcome of interest	Control for factor	Assessment of outcome	Follow-up long enough	Adequacy of follow-up	Total score
Fu et al. (2001)	*	*	*		**	*			6

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[22]									
Sun et al. (2001) [27]	*	*	*		**	*			6
Liu et al. (2002) [25]	*	*	*		**	*			6
Denti et al. (2004) [10]	*	*	*		**	*	*	*	8
Zhang et al. (2004) [29]	*	*	*		**	*			6
Johnsen et al. (2005) [12]	*	*	*		**	*			6
Geng et al. (2006) [23]	*	*	*		**	*	*		7
Peng et al. (2009) [26]	*	*	*		**	*			6
Chen et al. (2010) [21]	*	*	*		**	*	*		7

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Aberg et al. (2011) [14]	*	*	*		**	*	*	*	8
Dong et al. (2014) [11]	*	*	*		**	*			6
Tang et al. (2014) [13]	*	*	*		**	*	*	*	8
Zhang et al. (2016) [30]	*	*	*		**	*	*	*	8
Li et al. (2016) [24]	*	*	*		**	*			6
Armbrust et al. (2017) [19]	*	*	*		**	*	*	*	8
Tao et al. (2018) [28]	*	*	*		**	*	*	*	8
Aberg et al. (2019) [20]	*	*	*	*	**	*	*	*	9

*The study scored one point in the project; **The study scored two points in the project.

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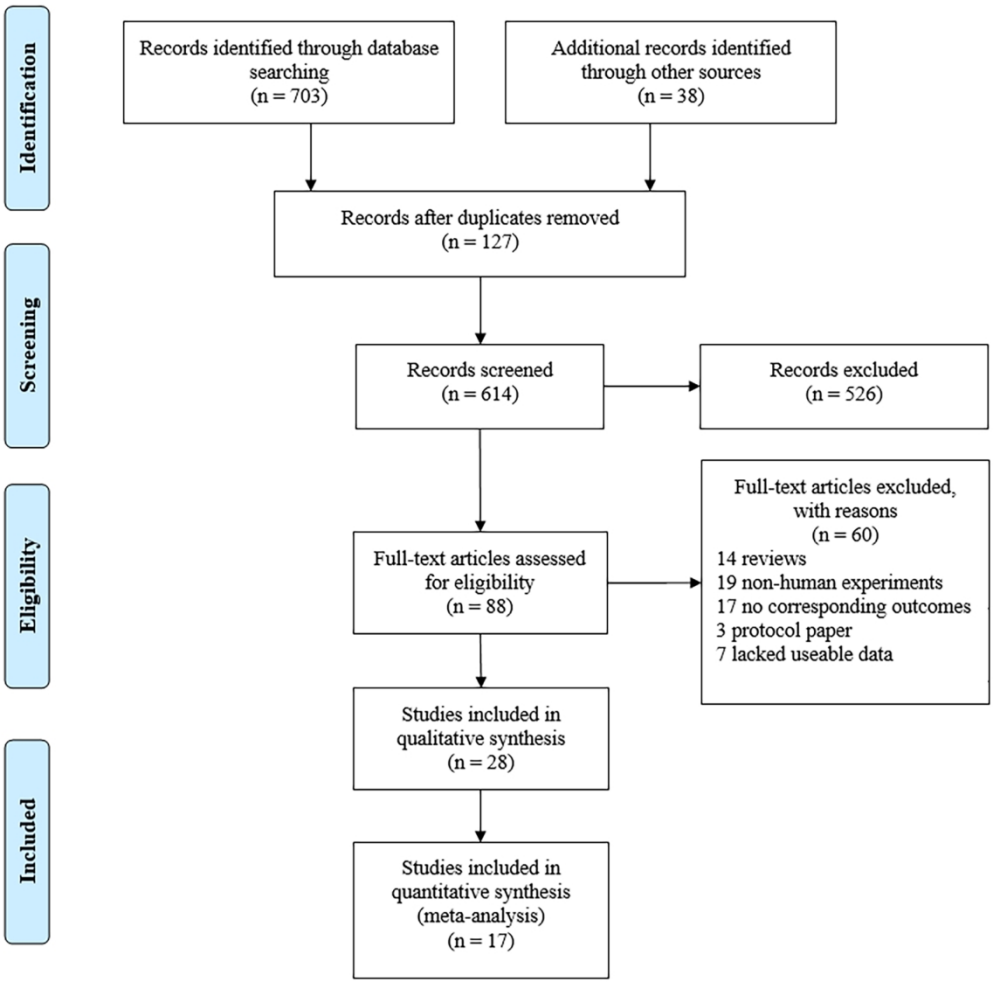


Figure 1

47x46mm (1000 x 1000 DPI)

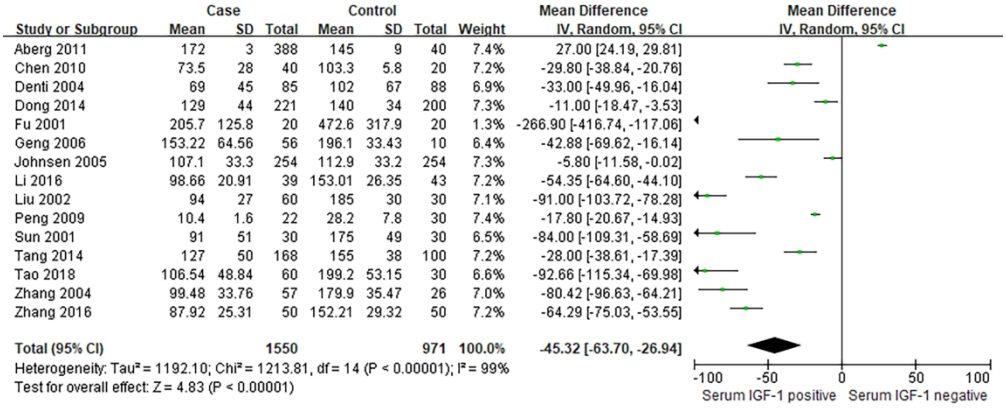


Figure 2

71x29mm (1000 x 1000 DPI)

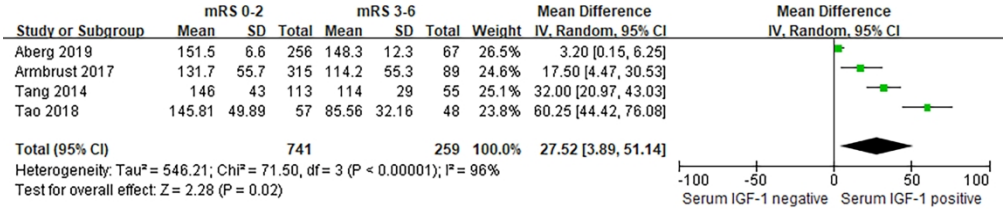


Figure 3

100x21mm (1000 x 1000 DPI)

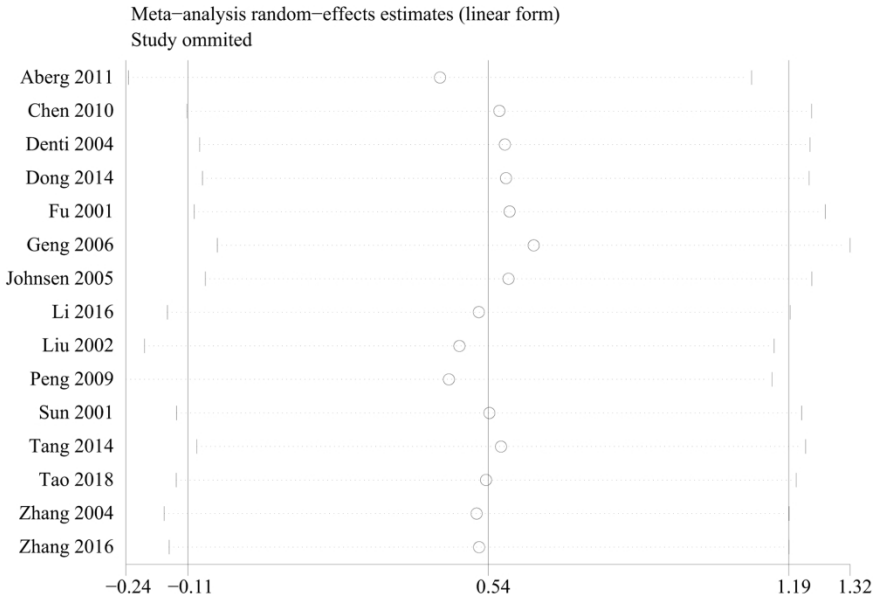


Figure 4

55x37mm (1000 x 1000 DPI)

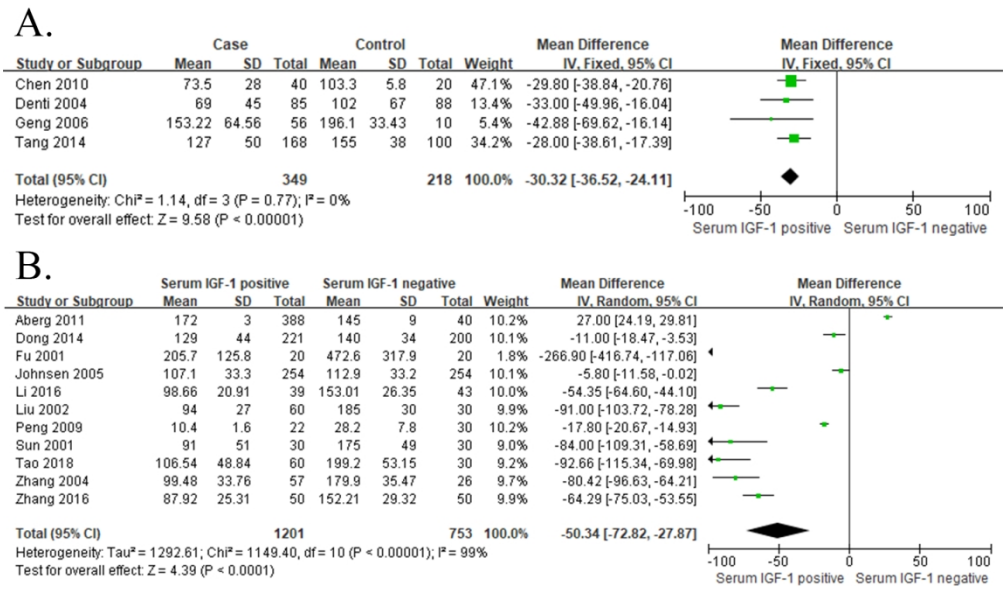


Figure 5

50x30mm (1000 x 1000 DPI)

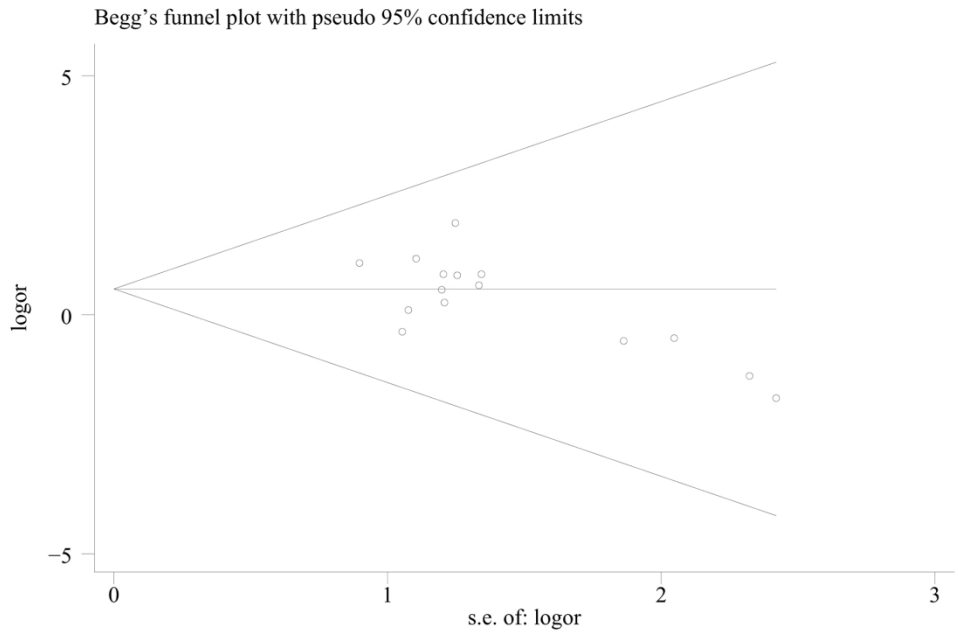

Begg's funnel plot with pseudo 95% confidence limits
5
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s.e. of: logor

Figure 6

Search strategy

This is a systematic review and meta-analysis about the relationship between serum insulin-like growth factor 1 levels and ischemic stroke. To ensure the rationality and extensiveness of data search, detailed database retrieval strategies for retrieval process will presented as followings.

Step1: Clarify retrieval problems and requirements

We transform the problem into the PICO model.

Step2: Clear search source

PubMed, Scopus, Embase, China National Knowledge Infrastructure (CNKI) and Web of Science were searched for studies published or presented on or before December 29, 2020. For maximum search, publications are not limited to English.

Step3: Determine search terms

According to the study content, we have identified "IGF-1" and "stroke" as search terms. To achieve maximum sensitivity of the search strategy and to include more extensive studies, Medical Subject Headings (MeSH) terms and the keywords that were retrieved in the fifield of Title/Abstract were used in combination.

After searching, we determined the subject words and free words as follows:

Subject words: Insulin-Like Growth Factor I

Free words: Insulin-Like Somatomedin Peptide I; Insulin Like Somatomedin Peptide I; Somatomedin C; IGF-I-SmC; IGF-1; IGF-I; Insulin Like Growth Factor I.

Subject words: Stroke

Free words: Strokes; Cerebrovascular Accident; Cerebrovascular Accidents; CVA (Cerebrovascular Accident); CVAs (Cerebrovascular Accident); Cerebrovascular Apoplexy; Apoplexy, Cerebrovascular; Vascular Accident, Brain; Brain Vascular Accident; Brain Vascular Accidents; Vascular Accidents, Brain; Cerebrovascular Stroke; Cerebrovascular Strokes; Stroke, Cerebrovascular; Strokes, Cerebrovascular; Apoplexy; Cerebral Stroke; Cerebral Strokes; Stroke, Cerebral; Strokes, Cerebral; Stroke, Acute; Acute Stroke; Acute Strokes; Strokes, Acute; Cerebrovascular Accident, Acute; Acute Cerebrovascular Accident; Acute Cerebrovascular Accidents; Cerebrovascular Accidents, Acute.

" IGF-1" was selected in the MeSH term and "Insulin-Like Somatomedin Peptide I", "Insulin Like Somatomedin Peptide I", "Somatomedin C", "IGF-I-SmC", "IGF-I", "Insulin Like Growth Factor I"

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were retrieved in the fifield of Title/Abstract. All the above were connected by “OR”. “Stroke” was selected in the MeSH term. Ultimately, “AND” was used to connect the two parts.

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	Title page, page 1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	Abstract, page 2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	Introduction, page 2, 3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	Introduction, page 2, 3
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	Study protocol, page 3
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	Eligibility criteria, page 3
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	Literature search strategy, page 3
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Literature search strategy, page 3
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	Literature search strategy, page 3
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	Data extraction and quality assessment, page 3
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	Data extraction and quality assessment, page 3
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	Data extraction and quality assessment, page 3
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	Statistical analysis, page 4
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	Statistical analysis, page 4

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	Statistical analysis, page 4
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	Statistical analysis, page 4
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	Study selection and characteristics, Figure 1, page 4
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	Study selection and characteristics, Table 1, page 4
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	Publication bias analysis and sensitivity analysis, Fig. 6, page 5
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	Fig. 2-5
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	Results, page 4, 5
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	Publication bias analysis and sensitivity analysis, Fig. 6, page 5
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Results, Table 3, Fig 4,5, page 4, 5
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	Discussion, page 6
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	Discussion, page 6, 7
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	Discussion, page 5-7
FUNDING			

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PRISMA 2009 Checklist

Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	Funding, page 7

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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Keywords:	Stroke < NEUROLOGY, Biochemistry < NATURAL SCIENCE DISCIPLINES, Adult neurology < NEUROLOGY

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The relationship between serum insulin-like growth factor 1 levels and ischemic stroke: a systematic review and meta-analysis

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Abbreviations

IGF-1: insulin-like growth factor 1; MD: mean difference; CI: confidence interval; IGFs: Insulin-like growth factors; NO: nitric oxide; CNKI: China National Knowledge Infrastructure; CT: computed tomography; MRI: magnetic resonance imaging; BMI: body mass index; NIHSS: National Institutes of Health Stroke Scale; mRS: modified Rankin Scale; NOS: Newcastle-Ottawa Scale; GH: Growth hormone; IGFBPs: insulin-like growth factor binding proteins; IGF-1R: insulin-like growth factor 1 receptor; CSF: cerebral spinal fluid; PRISMA: preferred reporting items for systematic reviews and meta-analyses

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Abstract

Objective: To assess the association of insulin-like growth factor 1 (IGF-1) with the risk of incident ischemic stroke and outcome after ischemic stroke.

Design: A systematic review of primary studies.

Setting: Hospitals in Western Sweden, Italy, China and Denmark.

Methods: A search was carried out in eligible studies in electronic databases (PubMed, Scopus, Embase, China National Knowledge Infrastructure and Web of Science) updated to December 29, 2020. The relevant data were extracted in order to conduct the meta-analysis. Review Manager version 5.2 was utilized to pool data and calculate the mean difference (MD) and its 95% confidence interval (CI). Heterogeneity, subgroup analysis, sensitivity analysis and publication bias were also performed in this meta-analysis.

Results: A total of 2277 patients were included in 17 studies. This meta-analysis indicated that higher serum IGF-1 levels were significantly correlated with less risk of ischemic stroke (MD=-45.32 95% CI = [-63.70, -26.94], P < 0.00001, I^2=99%) and better improvement of outcome after ischemic stroke (MD=27.52, 95% CI= [3.89, 51.14], P=0.02, I^2=96%). According to subgroup analysis, heterogeneity comes from country, sample size, male and the time from symptom onset to blood collection. Sensitivity analysis showed that there was no significant influence of any individual study on the pooled mean difference. The effect of high heterogeneity on result credibility was eliminated when four included studies were merged (MD=-30.32, 95% CI= [-36.52, -24.11], P< 0.00001, I^2=0%). Moreover, no potential publication bias was discovered in this meta-analysis.

Conclusion: Higher serum IGF-1 was significantly correlated with a lower risk of ischemic stroke. In view of the high degree of heterogeneity, it may need more studies to confirm the prognostic value of serum IGF-1 levels in ischemic stroke and explore the sources of heterogeneity.

Keywords: insulin-like growth factor-1, outcome, ischemic stroke, risk.

Strengths and limitations of this study

- This is a meta-analysis of 17 published studies comprising a total of 2277 patients in this study.
- The included studies were heterogeneous in the time from symptom onset to blood collection and follow-up duration at the end of the study.
- Study populations were limited to not less than 25 cases and no more than 1000 cases to prevent large samples from affecting the overall results.
- A thorough systematic review and meta-analysis provide confidence in the findings.
- Begg's test of publication bias was not performed on the relationship between serum IGF-1 and poststroke functional independence due to insufficient included studies.

Introduction

Insulin-like growth factors (IGFs), confirmed as peptide hormones, share similar structural homology with insulin. Insulin-like growth factor 1 (IGF-1) protects cells from apoptosis and functions as a vital regulator in cell survival [1, 2]. A growing body of evidence has established that IGF-1 influenced vascular function in many

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different ways, possessing anti-inflammatory properties and inducing vasodilation by promoting nitric oxide (NO) production [3, 4]. Recently, substantial evidence has accrued to demonstrate that IGF-1 plays a significant role in many cerebral diseases, for example, depression in ischemic stroke, cerebrovascular disease and Alzheimer's disease [5-7].

Stroke is a leading cause of death and disability worldwide, especially in the current aging society [8]. Animal studies and observational studies on humans have been trying to establish an association between IGF-1 and the risk of ischemic stroke. However, the association of IGF-1 with the risk of incident ischemic stroke still remains unclear. Seven previous studies are exploring the association manifested inconsistent conclusions. On the one hand, five studies showed that IGF-1 possessed a protective effect against ischemic stroke [9-13]. On the other hand, no association between incident ischemic stroke and IGF-1 was uncovered in two studies [14, 15]. Moreover, IGF-1 is widely disseminated and highly expressed in the nervous system [16], which exerts neuroprotective effects and accelerates proliferation of neurons. According to some studies, circulating serum IGF-1 is correlated with improvement of functional outcome after ischemic stroke and might predict outcome after ischemic stroke [13, 14, 17-19]. Actually, the outcome after ischemic stroke relies on complex interactions of multiple factors, and it still needs more studies to disclose whether the association is causative.

This meta-analysis was conducted in order to evaluate the association of IGF-1 with the risk of incident ischemic stroke and the outcome after ischemic stroke.

Methods

Patient and Public Involvement

This article didn't involve patients and or public.

Study protocol

This meta-analysis was carried out in accordance with the preferred reporting items for systematic reviews and meta-analyses (PRISMA) statement (Additional file 1).

Literature search strategy

PubMed, Scopus, Embase, China National Knowledge Infrastructure (CNKI) and Web of Science were searched for studies published or presented on or before December 29, 2020. Publications are not restricted to English. To achieve maximum sensitivity in the search strategy and to include more extensive studies, Medical Subject Headings (MeSH) terms and the keywords that were retrieved in the field of Title/Abstract were used in combination. "IGF-1" was selected in the MeSH term and "Insulin-Like Somatomedin Peptide I", "Insulin Like Somatomedin Peptide I", "Somatomedin C", "IGF-I-SmC", "IGF-I" and "Insulin Like Growth Factor I" were retrieved in the field of Title/Abstract. All the above are connected by "OR". "Stroke" was selected in the MeSH term. Ultimately, "AND" was utilized to connect the two parts. All relevant articles were found first, and then

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irrelevant articles were manually excluded through title and abstract screening and full-text assessment. To avoid selection bias，all work was carried out independently by two authors.

Eligibility criteria

Studies were included if they fulfilled the following criteria:

- Studies were confined to those involving human subjects.
- There were no restrictions on the information of language, year of the study, patient age, imaging criteria, body mass index (BMI) and the National Institutes of Health Stroke Scale (NIHSS) score.
- Study populations had no fewer than 25 cases and no more than 1000 cases.
- Study had a definite inclusion criterion for acute ischemic stroke.
- Patients who suffered from stroke were confirmed by computed tomography (CT) or magnetic resonance imaging (MRI) before treatment.
- Studies reported the complete outcomes.
- Study used a prospective or retrospective cohort design, or nested case–control design.
- If duplicate studies with an accumulating number of patients or something else were published, only the most complete studies were included in the analysis performed in this study.

Data extraction and quality assessment

All data were extracted independently from the tables and figures of included studies by two authors in an unblinded fashion or adjudicated by the third author in case of disagreement. Moreover, another author took responsibility for the accuracy and reliability of all of the extracted data. The modified Rankin Scale (mRS) is a clinical poststroke functional independence assessment tool. For each study, the study characteristics (author, publication year, study period, country), main characteristics of the patients (age gender and BMI) and outcomes (the number of ischemic stroke patients, the number of people in control, serum IGF-1 and mRS at 90 days) were extracted independently by two authors. The quality of the included studies was assessed by two other authors using the modified Newcastle-Ottawa Scale (NOS), while 6 points or more was considered to be of high quality.

Outcome measure

The Mean, SD, Total of serum IGF-1 (ng/mL) in case and control groups is a main indicator of stroke risk assessment. The mRS is an evaluation tool to assess the clinical poststroke functional independence (detailed grades of mRS: 0 - No symptoms; 1 - No significant disability; 2 - Slight disability; 3 -Moderate disability; 4 - Moderately severe disability; 5 - Severe disability; 6 - Dead). The prespecified primary outcome measures were the mRS score of 0 - 2 and the mRS score 3 - 6.

Statistical analyses

Statistical analysis was implemented using Review Manager version 5.2 (The Nordic Cochrane Centre, the

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Cochrane Collaboration, Copenhagen, Denmark). The combined effect of mean difference (MD) and 95% confidence interval (CI) was defined by a random-effect model or a fixed-effect model. The heterogeneity among included studies was tested using the inconsistency index (I^2). A random-effect model was used when there was a high level of heterogeneity (I^2 values > 50%). A P value <0.05 was considered statistically significant. For indicators with higher heterogeneity, sensitivity analysis and subgroup analyses were carried out to identify the reliability of the meta-analysis and meta-regression was used to explore the sources of heterogeneity. Publication bias was evaluated qualitatively by visual observation of funnel graphs and was assessed quantitatively by the Begg's test (Stata 12.0), which manifests a publication bias with P <0.05. The number, mean difference (MD), standard deviation (SD) of the experimental group and control group of the included studies and country, sample size, male, the time from symptom onset to blood collection of 17 included studies were imported into Stata 12.0 software.

Results

Study selection and characteristics

Detailed steps of the inclusion and exclusion criteria of the study are shown in Fig 1. Briefly, a total of 703 studies were retrieved through five electronic database searches. Thirty-eight additional studies were identified from the references of included studies after removing the duplicates. On the basis of extraneous titles and abstracts, 526 studies were excluded. Several studies were excluded because they were reviews, non-human experiments; did not present the corresponding outcomes and lack usable data. Ultimately, the final meta-analysis was conducted after a complete evaluation of 17 studies that met the inclusion criteria [10-14, 20-30].The basic characteristics and quality score of the included studies are presented in Table 1. The quality score of studies was proven to be acceptable (Table 2).

Fig. 1 Study search and selection flow diagram.

Relationship between serum IGF-1 and ischemic stroke risk

The relationship between serum IGF-1 and ischemic stroke patients was evaluated, and there was a high degree of heterogeneity among these included fifteen studies (I^2=99%). Thus, the random-effects model was utilized to pool MDs. The consequence manifested a significant difference (MD = -45.32, 95% CI = [-63.70, -26.94], P < 0.00001, I^2=99%, random effect) (Fig. 2).

Fig. 2 Forest plot of the relationship between serum IGF-1 and ischemic stroke risk.

Relationship between serum IGF-1 and poststroke functional independence

Poststroke functional independence was identified by mRS at 90 days. The relationship between serum IGF-1 and poststroke functional independence was evaluated, and the random-effects model was utilized to pool MDs. The result showed a significant difference (MD=27.52, 95% CI= [3.89, 51.14], P=0.02, I^2=96%, random effect)

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(Fig. 3).

Fig. 3 Forest plot of the relationship between serum IGF-1 and poststroke functional Independence.

Regression of heterogeneity sources

There is high heterogeneity in analyses so we use meta-regression to explore heterogeneity. From the analysis results, we can see that the sample size may be the primary source of heterogeneity (Table 3).

Table 3 Regression of heterogeneity sources

Item	Number of studies	95% CI	P value
Country	15	[-41.50, 78.27]	0.516
Sample size	15	[0.05, 0.28]	0.008
Male	14	[-4.42, 1.57]	0.320
The time from symptom onset to blood collection	14	[-18.71, 83.61]	0.192

Subgroup analysis of heterogeneity sources

In view of enormous heterogeneity, subgroup analysis of heterogeneity sources was performed to explore heterogeneity sources. The heterogeneity of each subgroup decreased modestly to varying degrees. Notably, the heterogeneity of the time from symptom onset to blood collection presented a significant decrease in parallel with other subgroup results (Table 4).

Table 4 Subgroup analysis of heterogeneity sources

Item	Subgroup	Number of studies	MD	95% CI	P value	Heterogeneity (I^2)
Country	Asian	12	-55.44	[-72.07, -38.82]	< 0.00001	97%
	Non-Asian	3	-2.92	[-32.42, 26.58]	0.85	99%
Sample size	≥100	6	-18.84	[-45.77, 8.08]	0.17	99%
	<100	9	-65.16	[-88.48, -41.83]	< 0.00001	97%
Male	≥60%	6	-65.89	[-107.51, -24.26]	0.02	98%
	<60%	8	-35.21	[-65.91, -4.52]	0.2	99%
The time from symptom	≤48 hours	7	-30.04	[-40.05, -20.03]	< 0.00001	86%
	>48 hours	4	-52.40	[-120.79, 15.99]	0.13	99%
	≤24 hours	5	-33.76	[-49.51, -18.01]	< 0.0001	89%

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onset to blood collection						

Abbreviations: MD, mean difference; CI, confidence interval.

Sensitivity analysis and adjusted relationship between serum IGF-1 and ischemic stroke

Sensitivity analysis was performed by omitting one study by turn, which showed that there was no significant influence of any individual study on the pooled MDs (Fig. 4). Moreover, in pursuit of a low degree of heterogeneity and to uncover low degree of heterogeneity sources, the included studies were randomly merged. It was discovered that heterogeneity reached its minimum when four included studies were merged (MD=-30.32, 95% CI= [-36.52, -24.11], P< 0.00001, I^2=0%) (Fig.5 A) [10, 13, 21, 23]. The other eleven included studies, in the other hand, remained high degree of heterogeneity (MD=-50.34, 95% CI= [-72.82, -27.87], P<0.00001, I^2=99%) (Fig. 5 B) [11, 12, 14, 22, 24-30]. Taken together, high degree of heterogeneity seemed not to discredit the final merged result.

Fig. 4 Forest plot in the sensitivity analysis of serum IGF-1 and ischemic stroke risk.

Fig. 5 Forest plot of the adjusted relationship between serum IGF-1 and ischemic stroke risk in (A) four included studies and (B) eleven included studies.

Publication bias analysis

According to Begg's test (P = 0.092 > 0.05) and the funnel plot (Fig. 6), no potential publication bias among the 17 included studies was noted in this meta-analysis.

Fig. 6 Begg's funnel plot for publication bias test of the relationship between serum IGF-1 and ischemic stroke risk.

Discussion

The mechanism of serum IGF-1 in evaluating the risk and prognosis of ischemic stroke

This meta-analysis merged 17 studies comprising 2277 patients to evaluate the association of serum IGF-1 with the risk of incident ischemic stroke and the prognostic value of serum IGF-1 in ischemic stroke. The results demonstrated that serum IGF-1 could be utilized to evaluate the risk of incident ischemic stroke (MD = -45.32, 95% CI = [-63.70, -26.94], P < 0.00001, I^2=99%, random-effect) and the outcome after ischemic stroke (MD=27.52, 95% CI= [3.89, 51.14], P=0.02, I^2=96%, random-effect). The main mechanism is as follows: Serum IGF-1 mainly comes from the liver, a small part comes from the kidneys and peripherals [31]. In body fluids, IGF-1 has a high affinity with IGF binding proteins (IGFBPs). IGFBPs regulate the biological activity and bioavailability of IGF-1, and the intake of serum IGF-1 enters the cerebrospinal fluid through the blood-brain

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barrier (BBB) [32]. IGF-1 binds to the widely distributed receptor IGF-1 receptor (IGF-1R) in the brain to activate the two main signal pathways of PI3K/Akt and MAPK, thereby inhibiting neuronal apoptosis [33], promoting the survival of oligodendrocytes [34], and protecting the hippocampus [35].

Discussion on the causes of heterogeneity

However, there was an inevitable high degree of heterogeneity among the included studies in this meta-analysis. In order to make the conclusion more persuasive and scientific, subgroup analysis was performed to explore the heterogeneity sources. As the results suggested, the time from symptom onset to blood collection appeared to be part of the source of heterogeneity, while heterogeneity decreased modestly to varying degrees. A study reported that stroke severity was related to decreased IGF-1 in rats as well as humans, which meant that the different stroke severity of recruited patients may also cause heterogeneity [11, 36]. Besides, insufficient follow-up duration may be the heterogeneity source. Due to the inability to obtain the original data, the heterogeneity from the time of symptom onset to blood collection, follow-up duration and stroke severity of recruited patients cannot be explored further. By scrutinizing the included studies, it was found that ischemic stroke patients who were recruited were subjected to the World Health Organization criteria in some included studies [10, 12, 13], whereas others included studies were not subjected to the criteria. In addition to different patient recruitment criteria, the nutritional status of the patient may cause heterogeneity, which can be indirectly reflected by BMI. Many ischemic stroke patients are followed by accelerated catabolism and decreased food intake, which results in a decrease in body muscle mass and BMI. The current study cannot rule out the possible effect of malnutrition on serum IGF-1. Moreover, forest plots in the sensitivity analysis (Fig. 4) and of the adjusted relationship between serum IGF-1 and ischemic stroke risk (Fig. 5) both demonstrate that the exclusion of any included study and a high degree of heterogeneity do not affect the final combined result.

The relationship between growth hormone and IGF-1 in cerebrovascular diseases

Growth hormone (GH) exerts much of its biological activity by promoting the production of IGF-1, and IGF-1 can also mediate the effects of GH in turn. The majority of circulating serum IGF-1 is completed with IGFBPs and acid labile subunits, which prevent IGF-1 from binding to the insulin receptor or modulate IGF-1's binding to the IGF-1R [37]. It is universally acknowledged that the bioavailability of IGF-1 is regulated by IGFBPs and that only unbound or free IGF-1 possesses biological activity. As part of the body's aging process, IGF-1 decreases sharply. Some human studies demonstrate that the effects of circulating serum IGF-1 on cerebrovascular events may be induced by metabolic syndrome [38, 39]. Furthermore, patients with growth hormone deficiency may have lower circulating IGF-1 levels, which raises the risk of cerebrovascular diseases significantly [40]. Notably, it was reversible with growth hormone replacement [41].

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Insufficiency of previous related research

Previous studies assessing the association of IGF-1 with the risk of incident ischemic stroke and the prognostic value of serum IGF-1 in ischemic stroke reported controversial findings. A study performed by Dong et al. (2014) and colleagues suggested that lower IGF-I levels are significantly related to risk of stroke and stroke severity, but IGFBPs levels were not obtained in that biologically active IGF-I was not fully represented [11]. Johnson et al. (2005) [12] found inverse associations between levels of IGF-I and IGFBP-3 and the risk of ischemic stroke. However, another study showed no association with incident ischemic stroke [15]. Meanwhile, Aberg et al. (2011) [14] suggested that serum IGF-1 levels were closely related to improvements in the mRS and Scandinavian Stroke Scale tested at 3 and 24 months. However, the widespread variation of sampling may have influenced conclusions. Mattlage et al. (2016) [42] reported the potential importance of the change in IGF-1 and IGF-1 ratio to IGFBP-3 during the first week of stroke in understanding recovery. The limitation of the study was that serum IGF-1 levels were not supplemented with the levels of IGF-1 in the cerebral spinal fluid (CSF). Because in ischemic stroke, IGF-1 transfers from the blood to the CSF. Nevertheless, another study showed an inconsistent conclusion that higher serum IGF-1 levels were modestly associated with better outcomes after ischemic stroke [20].

Strengths and limitations

To our knowledge, this meta-analysis is the first one to investigate the relationship between serum IGF-1 levels and ischemic stroke. Shaheen et al. (2018) [43] have found that serum IGF-1 was an independent risk factor for ischemic stroke in the Egyptian population which is consistent with our study. This research has geographic limitations so this meta-analysis included both English and Chinese studies so that publication bias was avoided to a great extent. Besides, the eligibility criteria that study populations were not less than 25 cases and not more than 1000 cases to prevent large samples from affecting the overall results. Aberg et al. (2019) [44] have proved that levels of serum IGF-1 are associated with near 3-month outcomes. However, the study only included 354 patients, maening that the sample size is too small. Otherwise, a comprehensive database search was performed in this meta-anlysis, and as a result, 17 studies comprising a total of 2277 patients were included. Despite the shining points of this meta-analysis, its limitations and shortcomings should be emphasized. A high degree of heterogeneity may cause the unreliability of the results. Meanwhile, Begg's test of publication bias was not performed on the relationship between serum IGF-1 and poststroke functional independence due to insufficient included studies. Ultimately, a data-feasible study [15] was excluded because of oversize cases, ambiguous patient recruitment criteria and no mention of stroke confirmed by imaging examination.

The clinical significance and future prospects of the study

Because the majority of serum IGF-1 binds to IGFBPs, the current measured serum IGF level cannot accurately represent the biological activity of IGF in circulation. It is hoped that the future meta-analysis will use

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the data obtained by the IGF-1 specific tyrosine kinase receptor activation analysis method (KIRA) [32] as much as possible. This method indirectly measures the actual biological activity by accurately assessing the activity of the official receptor. This meta-analysis significantly analyzed that higher IGF-1 correlated with a lower risk of ischemic stroke. It provides a basis for in-depth clinical research on the relationship between the level of IGF-1 and the neurological score [30] of stroke patients, post-stroke depression [45] dementia [46] and so on. In addition, it is also of great significance to the prevention of stroke (identification of high-risk groups and chemoprevention), early diagnosis, clinical treatment and prognosis.

Conclusions

This meta-analysis analyzed all the available interrelated information in the published studies and found that higher serum IGF-1 was significantly correlated with a lower risk of ischemic stroke. Concerning the relationship between serum IGF-1 and outcome after ischemic stroke, it may need more studies to confirm the prognostic value of serum IGF-1 levels.

Competing interests

None declared.

Ethics Approval

None declared

Additional file

Additional file 1: PRISMA checklist. (DOC)

Authors' contributions

YNL* and WNY* are co-first authors. YNL and WNY designed the study. WNY, YSZ and JHL performed the literature search. WNY, YNL and LCZ selected the studies for inclusion in the meta-analysis. SMC and YNL extracted the data. WNY performed the statistical analysis. YNL, WNY and JHL wrote the manuscript. LH and YYZ revised the manuscript. YYZ provided the funding. All authors read and approved the final manuscript.

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Availability of data and materials

All data analyzed during this study are included in this article.

Ethics approval and consent to participate

Not applicable.

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Consent for publication

Not applicable.

Competing interests for Reviewer

None declared

Acknowledgments

Not applicable

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Table 1. Basic characteristics of included studies in the meta-analysis

Study	Study period	Country	The time from symptom onset to blood collection	Age (y) [a or b]	Male (%)	BMI (kg/m^2) [a or b]	Follow-up duration	IGF-I (ng/mL)			
								Risk (Mean, SD or IQR, Total)		mRS (Mean, SD, Total)	
								Case	Control	mRS 0-2	mRS 3-6
Fu et al. (2001) [22]	Not Stated	China	NA	57-74	55.00	NA	NA	205.7, 125.8, 20	472.6, 317.9, 20	NA	NA
Sun et al. (2001) [27]	1998.9-1999.10	China	≤24 hours	67	70.00	26.45±2.51	NA	91, 51 ,30	175, 49 ,30	NA	NA
Liu et al. (2002) [25]	1999.2-2002.3	China	3 days	63.85±8.74	63.33	27±5	NA	94, 27, 60	185, 30, 30	NA	NA
Denti et al. (2004) [10]	1998.1-2000.6	Italy	≤24 hours	83±7	34.12	25±5	6 months	69, 45, 85	102, 67, 88	NA	NA
Zhang et al. (2004) [29]	2000.12-2001.6	China	NA	64±11	66.67	NA	NA	99.48, 33.76, 57	179.9, 35.47, 26	NA	NA
Johnsen et	1993.12-1997.5	Denmark	NA	50-64	61.00	26.5±4.0	NA	107.1,	112.9,	NA	NA

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al. (2005) [12]								33.3, 254	33.2, 254		
Geng et al. (2006) [23]	2003.6-2004.1	China	≤48 hours	65.08± 10.93	70.27	NA	NA	153.22, 64.56, 56	196.1, 33.43, 10	NA	NA
Peng et al. (2009) [26]	2006.12-2008.6	China	≤48 hours	NA	NA	NA	NA	10.4, 1.6, 22	28.2, 7.8, 30	NA	NA
Chen et al. (2010) [21]	2009.1-2009.12	China	≤24 hours	65.64±7.59	55	NA	NA	73.5, 28 ,40	103.3, 5.8, 20	NA	NA
Aberg et al. (2011) [14]	1998.8-2003.12	Western Sweden	4 days	55.5±0.86	56.65	26.2±0.43	3 months	172, 3, 388	145, 9, 40	NA	NA
Dong et al. (2014) [11]	2011.8-2013.8	China	≤24 hours	66 (57–77)	58.75	23.9(22.6–26.8)	NA	129, 44, 221	140, 34, 200	NA	NA
Tang et al. (2014) [13]	2012.8-2013.8	China	≤24 hours	72 (62–83)	53.57	25.3(21.8–28.4)	3 months	127, 50, 168	155, 38, 100	114, 29, 113	146, 43, 55
Zhang et al. (2016) [30]	2013.2-2015.9	China	NA	65.2±8.6	56.00	26.28+1.59	3 months	87.92, 25.31, 50	152.21, 29.32, 50	NA	NA

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Li et al. (2016) [24]	2014.5-2015.6	China	3 days	51.9±9.6	56.41	23.9±6.1	NA	98.66, 20.91, 39	153.01, 26.35, 43	NA	NA
Armbrust et al. (2017) [19]	NA	NA	≤24 hours	72.9±9.5	55.1	26.8±4.4	3 months	NA	NA	131.7, 55.7, 315	114.2, 55.3, 89
Tao et al. (2018) [28]	2016.1-2017.3	China	3 days	60.30±9.75	66.67	NA	3 months	106.54, 48.84, 60	199.2, 53.15, 30	145.81, 49.89, 57	85.56, 32.16, 48
Aberg et al. (2019) [20]	1998-2003	Western Sweden	≤24 hours	55.3±11	65.12	NA	7 years	NA	NA	151.5, 6.6, 256	148.3, 12.3, 67

[a] Median (IQR); [b] Mean (SD).

Abbreviations: BMI, body mass index; NA, not available; IQR, interquartile range; SD, standard deviation.

Table 2 Newcastle-Ottawa Scale for quality assessment

Study	Exposed cohort	Non-exposed cohort	Ascertainment of exposure	Outcome of interest	Control for factor	Assessment of outcome	Follow-up long enough	Adequacy of follow-up	Total score
Fu et al. (2001)	*	*	*		**	*			6

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[22]									
Sun et al. (2001) [27]	*	*	*		**	*			6
Liu et al. (2002) [25]	*	*	*		**	*			6
Denti et al. (2004) [10]	*	*	*		**	*	*	*	8
Zhang et al. (2004) [29]	*	*	*		**	*			6
Johnsen et al. (2005) [12]	*	*	*		**	*			6
Geng et al. (2006) [23]	*	*	*		**	*	*		7
Peng et al. (2009) [26]	*	*	*		**	*			6
Chen et al. (2010) [21]	*	*	*		**	*	*		7

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Aberg et al. (2011) [14]	*	*	*		**	*	*	*	8
Dong et al. (2014) [11]	*	*	*		**	*			6
Tang et al. (2014) [13]	*	*	*		**	*	*	*	8
Zhang et al. (2016) [30]	*	*	*		**	*	*	*	8
Li et al. (2016) [24]	*	*	*		**	*			6
Armbrust et al. (2017) [19]	*	*	*		**	*	*	*	8
Tao et al. (2018) [28]	*	*	*		**	*	*	*	8
Aberg et al. (2019) [20]	*	*	*	*	**	*	*	*	9

*The study scored one point in the project; **The study scored two points in the project.

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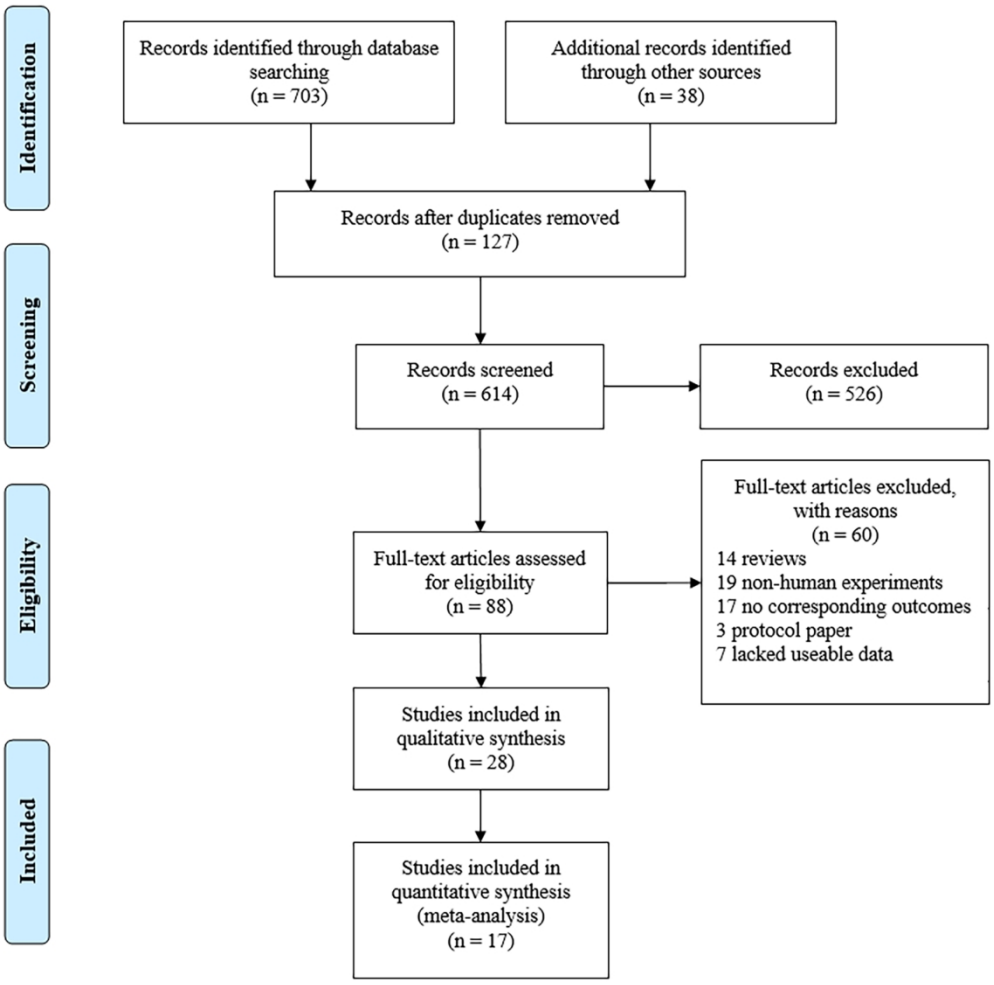


Figure 1

47x46mm (1000 x 1000 DPI)

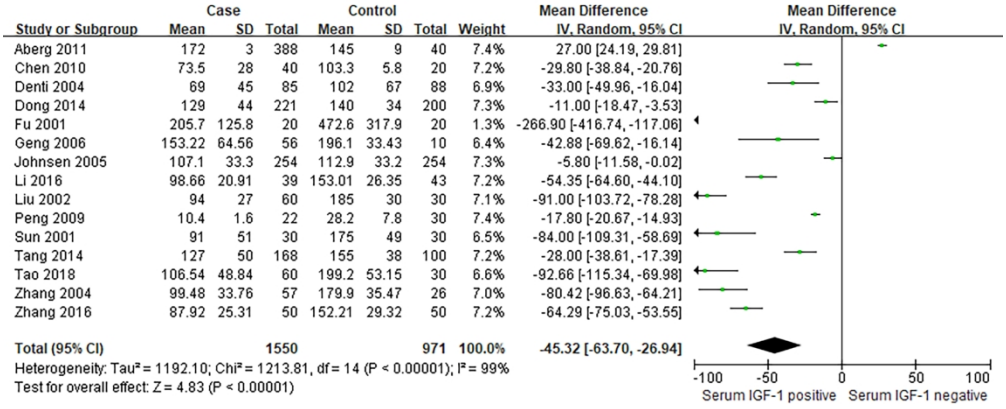


Figure 2

71x29mm (1000 x 1000 DPI)

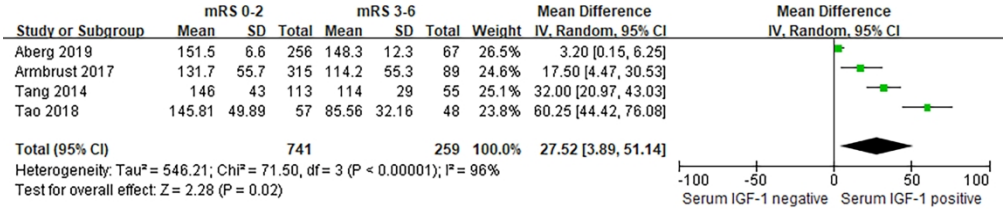


Figure 3

100x21mm (1000 x 1000 DPI)

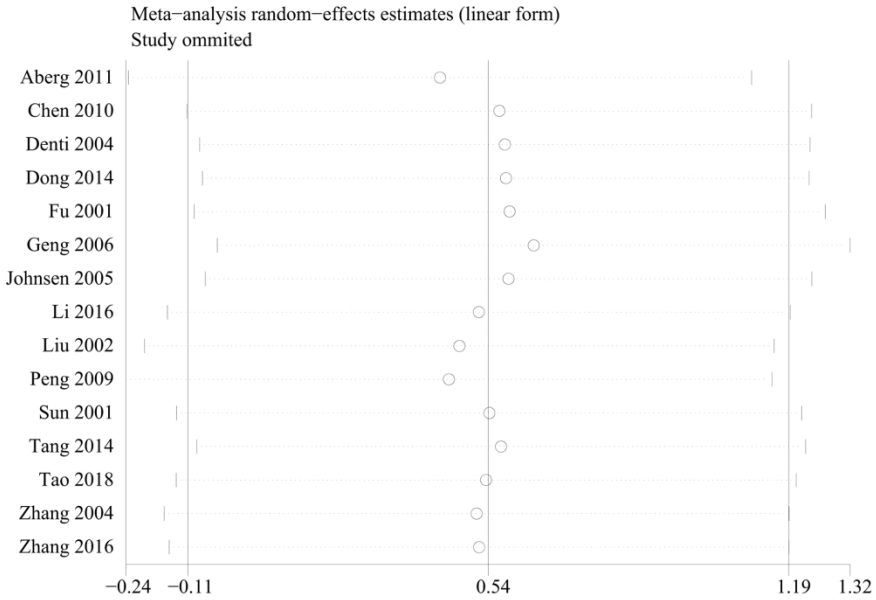


Figure 4

55x37mm (1000 x 1000 DPI)

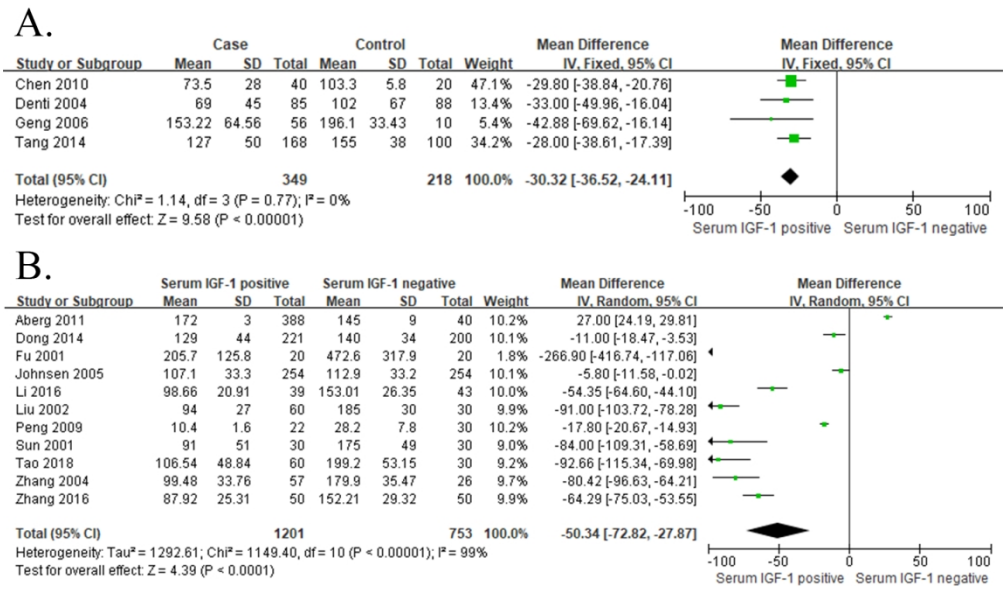


Figure 5

50x30mm (1000 x 1000 DPI)

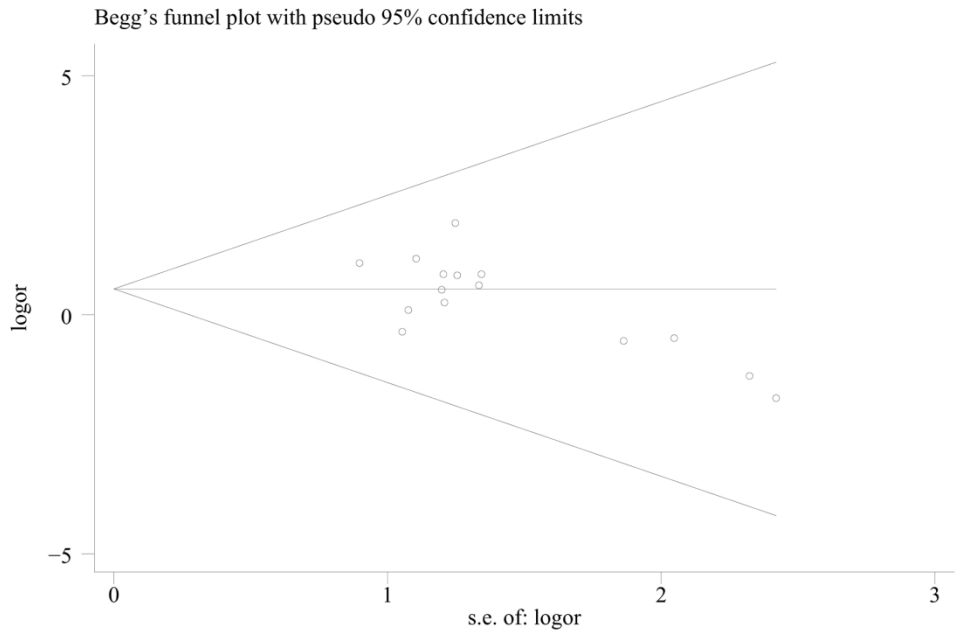

Begg's funnel plot with pseudo 95% confidence limits
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Figure 6

PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	Title page, page 1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	Abstract, page 2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	Introduction, page 2, 3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	Introduction, page 2, 3
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	Study protocol, page 3
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	Eligibility criteria, page 3
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	Literature search strategy, page 3
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Literature search strategy, page 3
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	Literature search strategy, page 3
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	Data extraction and quality assessment, page 3
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	Data extraction and quality assessment, page 3
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	Data extraction and quality assessment, page 3
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	Statistical analysis, page 4
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	Statistical analysis, page 4

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	Statistical analysis, page 4
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	Statistical analysis, page 4
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	Study selection and characteristics, Figure 1, page 4
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	Study selection and characteristics, Table 1, page 4
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	Publication bias analysis and sensitivity analysis, Fig. 6, page 5
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	Fig. 2-5
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	Results, page 4, 5
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	Publication bias analysis and sensitivity analysis, Fig. 6, page 5
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Results, Table 3, Fig 4,5, page 4, 5
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	Discussion, page 6
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	Discussion, page 6, 7
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	Discussion, page 5-7
FUNDING			

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Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	Funding, page 7

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